



U.S. ARMY

**U.S. ARMY GARRISON BENELUX
PERFORMANCE WORK STATEMENT
HVAC MAINTENANCE SERVICES**

PERFORMANCE WORK STATEMENT (PWS)

HVAC Services for Joint Forces Command (JFC) Brunssum – Buildings H-310 & H-602 (CYSS), USAG Benelux Brunssum Support Site, and Army Preposition Stock-2 (APS-2) Site Egelshoven

Dated: 17 February 2026

DRAFT

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PART 1 GENERAL INFORMATION

1 General Information:

This is a non-personal service(s) Performance Work Statement (PWS) under which the personnel rendering the service(s) are not subject, either by the contract's terms or by the manner of its administration, to the supervision and control usually prevailing in relationships between the Government and its employees.

1.1 Description of Services/Introduction:

This is a non-personal services PWS to provide preventive maintenance (PM), inventory, testing and labeling of HVAC equipment items and systems in facilities located at UBBS, CYSS and APS-2. The Government shall not exercise any supervision or control over contract service providers who are subcontractors performing services herein. Such contract service providers shall be accountable solely to the contractor who, in turn is responsible to the Government. The Contractor shall perform to the standards specified in this PWS. The Contractor shall provide all personnel, equipment, supplies, facilities, transportation, tools, materials, supervision, and other items and non-personal services necessary as defined in this PWS.

1.2 Background:

The USAG Benelux mission is to provide base operations to joint military, civilians and their families, to include interagency community members with a quality of life equal to the quality of their service. As part of the aforementioned base operational support, Preventative Maintenance, Ventilation, Air Conditioning, and Combination Systems (HVAC-LBK) are required to provide comfortable and safe facility conditions.

1.3 Scope:

The Contractor shall provide non-personal services for Heating, Ventilation, Air Conditioning, and Combination Systems (HVAC-LBK) on United States (U.S.) Army Installations in the Netherlands. These services include preventive maintenance, inventory, and labeling of all components, devices, and equipment related to HVAC-LBK. System components include, but are not limited to: indirect units, direct units, split units, electric and gas boilers, furnaces, compressors, control systems, burner assemblies, temperature controls, air curtains, air filters, condenser units, exhaust fans, ventilation fans, and recirculation units.

The scope of this contract includes all HVAC-related systems and associated components as identified in Technical Exhibit C, including but not limited to heating and combustion systems, hydronic heating and cooling systems, refrigeration and air-conditioning systems, stand-alone dehumidifiers, ventilation and air-side systems, fuel and gas distribution piping, control air and compressed air systems, controls and instrumentation, and all required preventive and corrective maintenance, inspections, testing, documentation, and environmental compliance activities associated with these systems.

1.4 Objectives:

The objective of this PWS is to provide HVAC scheduled preventive maintenance at facilities on the UBBS, CYSS and APS-E systems and equipment listed in this PWS on a recurring basis.

1.5 General Information

1.5.1 Quality Control Plan (QCP):

The Contractor shall develop and maintain a QCP to ensure services are performed in accordance with (IAW) this PWS. The Contractor shall develop and implement procedures to identify, prevent, and ensure non-recurrence of defective services. The Contractor's QCP is the means by which it assures that the work provided complies with the requirements of the contract.

The Contractor's Proposed QCP shall be submitted to the Contracting Officer (KO) through the Contracting Officer's Representative (COR) for review within thirty (30) days after date of contract award (*Deliverable 1*).

The Government will review and either notify the Contractor in writing of acceptance of the plan or return their comments to the Contractor within fifteen (15) days. If the Government has provided comments, the Contractor shall then have three (3) days to submit a Final QCP. After receipt of the Final QCP, the Contractor may receive the Contracting Officer's acceptance in writing. Any proposed changes to the accepted QCP are required to be resubmitted for acceptance by the Contracting Officer no later than thirty 30 days prior to the anticipated change and before implementation by the Contractor. The timeline noted above will apply for review and acceptance for proposed changes. At a minimum, the QCP must include and answer the following to be acceptable:

- A. A chart showing the organizational structure and lines of authority, the names, qualifications, duties, responsibilities, and classification of each member of the Contractor's Quality Control Team.
- B. How will the Contractor monitor work to ensure that performance complies with all deliverables (etc. timelines, deadlines, and goals)?
- C. How will the Contractor monitor work to ensure that performance complies with all specifications and requirements of the contract, including clauses?
- D. How will the Contractor monitor and ensure staff qualifications remain current and valid throughout contract performance?
- E. How will the Contractor ensure all keys issued will remain controlled items (Paragraph 1.7 Key Control)?
- F. How will the Contractor inventory and track maintenance of all Government Provided Equipment / Materials?
- G. How will the Contractor identify, investigate, and correct any non-conforming performance and prevent similar deficiencies in the future?
- H. How will the Contractor file and save all Quality Control related documents for the life of the contract plus 5 years?

1.5.1.1 Analysis:

An Analysis of Quality Control and Timeliness of Submittals (i.e. deliverables) shall be provided to the COR within thirty 30 days upon completion of each option period or Task Order (*Deliverable 2*).

1.5.1.2 Inform:

The Contractor shall inform the COR of issues or potential issues that might affect the performance within four (4) hours of identifying those issues. Verbal reports shall be followed up by written reports within three (3) work days. (*Deliverable 3*).

1.5.2 Quality Assurance:

The Government will evaluate the Contractor's performance under this PWS in accordance with the Quality Assurance Surveillance Plan (QASP). This plan is primarily focused on what the Government must do to ensure that the Contractor has performed in accordance with the performance standards. It defines how the performance standards will be applied, the frequency of surveillance, and the acceptable quality levels (performance thresholds).

1.5.3 Recognized Holidays:

The following provides information on recognized holidays for the purpose of the PWS. If submittal of any documentation (e.g. deliverables, submittals, etc.) deadlines fall on a holiday, the closest work day prior to the holiday will apply as the deadline for submittal.

1.5.3.1 U.S. Holidays:

Work shall be performed on U.S. holidays occurring during the normal workweek unless otherwise directed by the Contracting Officer. When a U.S holiday occurs on a Saturday or a Sunday, the holiday is observed on the preceding Friday or following Monday, respectively. [The contractor is required to perform services on holidays]

NAME OF U.S. HOLIDAY	DATE
New Year Day	January 1st
Martin Luther King Memorial Day	3rd Monday in January
President Day	3rd Monday in February
Juneteenth	June 19th
Memorial Day	Last Monday in May
Independence Day	July 4th
Labor Day	1st Monday in September
Columbus Day	2nd Monday in October
Veteran's Day	November 11th
Thanksgiving Day	4th Thursday in November
Christmas Day	December 25th

1.5.3.2 Host Nation Holidays:

Work shall not be performed on local Host Nation holidays occurring during the normal work week unless otherwise directed by the Contracting Officer.

Dutch Holidays:

NAME OF DUTCH HOLIDAY	DATE
New Year Day	January 1st
Easter Monday	Varies Annually
King's Day	April 27
Liberation Day	May 5 th
Ascension Day	Varies Annually
Whit Monday	Varies Annually
1 st Christmas Day	December 25th
Boxing Day	December 26th

1.5.4 Operating Hours:

The Government facility office hours, facility operating hours, and the Contractor support hour requirements often coincide; however, they may differ. Please refer to the following.

1.5.4.1 Government Facility Office Hours:

The Government facility office hours are 0800-1600 except U.S. Holidays identified in paragraph 1.5.3.1 above or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings. When an unforeseen installation closure occurs on a regularly scheduled day of work, the Contractor shall: Reschedule the work to be performed the following day unless the following day is a Saturday, Sunday, a legal public holiday, and when routine work is not scheduled for that day; or reschedule the work on any day mutually satisfactory.

1.5.4.2 Government Facility Operating Hours:

The Government facility operating hours occur twenty-four (24) hours a day /seven (7) days a week. The Government facility is open for operations during the recognized U.S. Holidays identified in paragraph 1.5.3.1 above. The Government facility may close due to local or national emergencies, administrative closings, or similar Government

directed facility closings and will be communicated by the Contracting Officer to the Contractor. Exceptional maintenance work at CYSS should be planned on days the location is closed or in the weekend due to child safety measurements. For other than firm fixed price contracts, the contractor will not be reimbursed when the government facility is closed for the above reasons. The Contractor must at all times maintain an adequate workforce for the uninterrupted performance of all tasks defined within this PWS when the Government facility is not closed for the above reasons. When hiring personnel, the Contractor shall keep in mind that the stability and continuity of the workforce are essential.

1.5.4.3 Contractor Support Hours:

The Contractor shall provide Maintenance support between the hours of 0800 to 1600, Monday through Friday except Host Nation Holidays identified in paragraph 1.5.3.2 above or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings. The Contractor shall at all times maintain an adequate workforce for the uninterrupted performance of all tasks defined within this PWS when the Government facility is not closed for the above reasons. When hiring personnel, the Contractor shall keep in mind that the stability and continuity of the workforce are essential.

1.5.4.4 Contractor Work Plan:

The Contractor shall prepare and maintain a Work Plan. The plan, as a minimum, shall identify regularly scheduled work (to include location, work to be completed, timeline, etc.) and, if applicable, how the Contractor intends to address and meet increased operations tempo (e.g. Contractor's need for overtime or extended work hours). If the need for overtime is anticipated, the Contractor must receive approval from the Contracting Officer at a minimum seven (7) calendar days prior to execution.) The Contractor's proposed Work Plan shall be submitted to the Contracting Officer via the COR for review within ten (10) days prior to the execution of the work. (Deliverable 4). The Government will review and either accept or return their comments to the Contractor within three (3) days). The Contractor shall then have three (3) days to submit a Final Work Plan. Any proposed change to the accepted Final Work Plan is required to be resubmitted via the COR for review and acceptance by the Contracting Officer at a minimum thirty (30) days prior to implementation. The timeline noted above shall apply for review and acceptance for proposed changes.

1.5.5 Training Holidays:

USAREUR has historically granted military personnel training holidays as compensation for time that they spent in field training. These are typically combined with regular holiday leave schedules (for example, Christmas half days or day on day off schedules). Training holidays may affect access to military operated facilities. If scheduled work or access to facilities is required, coordination with occupants will be necessary or the work in affected facilities shall be rescheduled after the training holiday. However, announced training holidays or exodus of military personnel may provide the Contractor with an opportunity for unrestricted access to facilities and areas. The Contractor shall take full advantage of such opportunities by planning maintenance or actions that are better accomplished when such areas are vacant.

1.5.6 Place of Performance:

The work to be performed under this PWS will be performed on the following installations:

JFC NATO (H-310 & H-602) Rimburgerweg 30 6445 PA Brunssum	USAG Brunssum Kranenpool 3 6443 VA Brunssum	APS Eysgelshoven Rimburgerweg 50 6471 XX Eysgelshoven
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1.6 Security Requirements:

The following information is provided on security related matters.

1.6.1 Security Program:

Contractor personnel performing work under this PWS must have a Dutch “Verklaring Omtrent het Gedrag” (VOG) (uitreksel uit het Basisregistratie Personen (BRP))” (Statement of Good Conduct issued by Justis Agency of the Netherlands Ministry of Justice and Security: <https://www.rijksoverheid.nl/wetten-en-regelingen/productbeschrijvingen/aanvragen-verklaring-omtrent-het-gedrag-vog-bij-justis>) within thirty (30) days after contract award, and must maintain the level of security required for the life of the contract.

1.6.2 Personnel Security Clearance Requirements:

RESERVED

1.6.3 Installation Access:

Access to U.S. installations and controlled areas is limited to personnel who meet security criteria and are authorized by Host Nation law to work in that country. Failure to submit required information/data and obtain required documentation or clearances in accordance with Army in Europe (AE) Regulation 190-16 (AE Reg 190-16) Installation Access Control, will be grounds for denying access to U.S. installations and controlled areas. The Contractor shall ensure that any subcontractors used in performance of this PWS complies with these requirements and that all employees, of both the Contractor and any subcontractor utilized by the Contractor, are made aware of and comply with these requirements. Should anything in this paragraph conflict with AE Reg 190-16 or (AE) Regulation 604-1, the regulation takes precedent.

The Contractor shall be aware of and comply with the requirements associated with Installation Access Control. The Government is not liable for any costs associated with performance delays due solely to a firm’s failure to comply with Installation Access Control System (IACS) processing requirements.

The Contractor shall return installation passes to the issuing IACS office when the contract is completed or when a Contractor employee no longer requires access.

AE Reg 190-16 can be found on the following website: <https://media.defense.gov/2025/Mar/13/2003666299/-1/-1/0/AEF190-16A.pdf>

1.6.3.1 Installation Pass Log Administration:

The Contractor shall provide the Contracting Officer, via the COR, an Initial Installation Pass Log within three (3) calendar days after contract award (*Deliverable 5*). The log must at a minimum contain the following information: full name, address, and DOB. The log shall be updated as employees are added or removed. The Updated Installation Pass Log shall be current and available at any time upon request by the Government and submitted to the COR within five (5) days after any changes to the log (*Deliverable 5*).

1.6.3.2 Collection Upon Employee Termination:

The Contractor shall collect the installation access passes the same day employment of an individual has expired or has been terminated and shall return them to the issuing office within three (3) days (*Deliverable 6*).

1.6.3.3 Collection Upon Contract Termination:

Upon termination or cancellation of the contract, the Contractor shall collect all outstanding installation access passes and return them to the issuing office within three (3) days (*Deliverable 6*).

1.6.3.4 Common Access Cards (CAC):

RESERVED

1.6.4 Background Checks:

The contractor must adhere to the requirements outlined in paragraph 1.6 above.

1.6.4.1 U.S. Citizen:

RESERVED

1.6.4.2 Local Nationals and Third Country Nationals:

RESERVED

1.6.5 Physical Security:

The contractor shall be responsible for safeguarding all government equipment, information and property provided for contractor use.

1.6.6 Operations Security (OPSEC) Requirements:

Contractor personnel shall adhere to facility security policies and restrictions. The Contractor shall immediately report suspicious activities to security personnel. Government issued access badges shall not be worn outside designated facilities where visible to the general public.

1.7 Key Control:

The Contractor shall establish and implement methods of making sure all keys issued to the Contractor by the Government are not lost or misplaced and are not used by unauthorized persons. NOTE: All references to keys include key cards. No keys issued to the Contractor by the Government shall be duplicated. The Contractor shall develop procedures covering key control that shall be included in the Contractor's Quality Control Plan. Such procedures shall include turn-in of any issued keys by personnel who no longer require access to locked areas. The Contractor shall immediately report any incidents of lost or duplicate keys to the Contracting Officer. In the event keys, other than master keys, are lost or duplicated, the Contractor shall, upon direction of the Contracting Officer, re-key or replace the affected lock or locks; however, the Government, at its option, may replace the affected lock or locks or perform re-keying. When the replacement of locks or re-keying is performed by the Government, the total cost of re-keying or the replacement of the lock or locks shall be deducted from payment due to the Contractor. In the event a master key is lost or duplicated, all locks and keys for that system shall be replaced by the Government and the total cost deducted from the payment due to the Contractor. The Contractor shall prohibit the use of Government issued keys by any persons other than the Contractor's employees. The Contractor shall prohibit the opening of locked areas by Contractor employees to permit entrance of persons other than Contractor employees engaged in the performance of assigned work in those areas, or personnel authorized entrance by the Contracting Officer. The Contractor shall establish and implement methods of ensuring that all lock combinations are not revealed to unauthorized persons. The Contractor shall ensure that lock combinations are changed when personnel having access to the combinations no longer have a need to know such combinations. These procedures shall be included in the Quality Control Plan.

1.8 Post Award Conference/Periodic Progress Meetings:

The Contractor shall attend any post award conference convened by the contracting activity or contract administration office in accordance with FAR Subpart 42.5. The Contracting Officer, COR, and other Government personnel, as appropriate, may meet periodically with the Contractor to review the Contractor's performance. At these meetings the Contracting Officer will apprise the Contractor of how the Government views the Contractor's performance and the Contractor shall apprise the Government of problems, if any, being experienced. Appropriate action shall be taken to resolve outstanding issues. These meetings shall be at no additional cost to the Government.

1.8.1 Attendance at Meeting and Conferences:

The Contractor shall attend, participate in, and furnish input to scheduled and unscheduled meetings, conferences, and briefings that relate to the functions and services herein as required by the Government to provide effective communication and impart necessary information. The Contract Manager or designated representative shall attend meetings as requested by the Government. Meeting attendees shall at times include Contractor managerial, supervisory, and other personnel knowledgeable of the subject matter. Meetings may start or end outside of regular duty hours.

1.9 Contracting Officer's Representative (COR):

Refer to Part 2 of this PWS for the definition of a COR. As determined by the Contracting Officer, a COR will be appointed and identified by letter of designation, a copy of which will be provided to the Contractor by the Contracting Officer. The designation letter states the responsibilities and limitations of the COR, especially with regard to changes in cost or price, estimates, or changes in delivery dates. The COR is not authorized to change any of the terms and conditions of the resulting order. The COR monitors all technical aspects of the contract and assists in contract administration. The COR is not authorized to obligate the Government. If the work is not written in the contract, the COR is not authorized to request new work. The COR is authorized to perform the following functions: assure that the Contractor performs the technical requirements of the contract, perform inspections necessary in connection with contract performance, maintain written and oral communications with the Contractor concerning technical aspects of the contract, issue written interpretations of technical requirements, including Government drawings, designs, specifications, monitor Contractor's performance and notifies both the Contracting Officer and Contractor of any deficiencies, coordinate availability of Government property, and coordinate site entry of Contractor personnel.

1.10 Key Personnel:

The following personnel are considered key personnel by the Government: Program/Contract Manager and Alternate Contract Manager. The Contractor shall provide a Program/Contract Manager who shall be responsible for the performance of the work. The name of this person and an Alternate Program/Contract Manager, who shall act for the Contractor when the Program/Contract Manager is absent, along with a Curriculum Vitae (CV) for each of these individuals, shall be provided in writing to the Contracting Officer no later than during the post-award meeting. (*Deliverable 7*). Acceptance of the CV by the Contracting Officer is required before these individuals are permitted to perform any task on the contract. The Contract Manager and Alternate Contract Manager shall have full authority to act for the Contractor on all contract matters relating to daily operation of the contract. The Contract Manager or Alternate Contract Manager shall be available 24-hours. Qualifications for all key personnel are listed below:

a. Contract Manager and Alternate Contract Manager: The Contract Manager and Alternate Contract Manager shall have full authority to act for the Contractor on all contract matters relating to daily operation of the contract. They shall fluently speak, read and write English and Dutch. The Contract Manager and Alternate Contract Manager shall have within the past ten (10) years, three (3) years' experience working as a Contract/Project manager consisting of a minimum of two (2) years manager and/or supervisory experience in managing facility/equipment maintenance and repair contracts.

b. Special Qualifications for Contractor/Subcontractor firm and employees: The Contractor and any subcontractors shall be fully certified, trained, and compliant with all applicable Dutch laws, regulations, and certification schemes governing the installations and systems covered under this contract. The Contractor shall ensure that all legally required inspections, certifications, and re-inspections are planned and performed within the prescribed regulatory timeframes and that all installations remain continuously compliant throughout contract performance.

For all work involving refrigeration and air-conditioning systems containing fluorinated greenhouse gases (F-gases), the following certifications are required:

1. BRL 100 – F-gas Company Certification, required for companies performing installation, maintenance, repair, leak checking, decommissioning, or recovery of F-gas-containing equipment.
2. BRL 200 – F-gas Personal Certification, required for individual technicians performing work on refrigeration or air-conditioning equipment.
3. Valid VCA certificate for all personnel performing services under this contract.

For combustion-based heating systems (boilers, burners, and other combustion equipment) subject to Dutch environmental legislation, all required inspections, maintenance, and testing shall be performed by a SCIOS-certified company using SCIOS-certified technicians. Applicable SCIOS scopes include, but are not limited to:

- Scope 1 / 2 – First Special Inspection (EBI) and Periodic Maintenance (PO)
- Scope 6 – Periodic Inspection (PI)
- Scope 7a – Gas safety inspection and integrity (tightness) testing of gas distribution piping
- Scope 7b – Inspection, maintenance, and technical management of fuel distribution piping systems
- Scope 7c – Emission and efficiency measurements (where required by environmental permits)
- Scope 14 – Inspection, maintenance, and certification of geothermal ground energy systems, where applicable.

All required certificates and inspection reports shall be provided to the KO and COR.

c. Technicians: Technicians shall read, write, and speak English and Dutch. The Contractor shall ensure that all employee certifications remain valid throughout the performance period of the contract. The Contractor shall provide copies of all required certifications to the KO and COR at the Post Award meeting and shall provide proof of certification for any replacement or newly assigned employee within five (5) workdays (*Deliverable 8*).

Technicians performing refrigeration or air-conditioning services involving F-gases shall hold:

1. BRL 200 – F-gas Personal Certification
2. A valid VCA certificate

Technicians performing inspections, maintenance, or testing on combustion installations or gas-fed heating systems shall hold the relevant SCIOS personal certifications for:

- Scope 1 / 2 (EBI/PO)
- Scope 6 (PI)
- Scope 7a (gas distribution piping)
- Scope 7b (fuel distribution piping)
- Scope 7c (emission and efficiency testing), where applicable
- Scope 14 (geothermal ground energy systems), where applicable

The Contractor shall ensure that only appropriately certified personnel perform work under the scopes for which they are qualified.

1.11 Identification of Contractor Employees:

All Contractor personnel attending meetings, answering Government telephones, and working in other situations where their contractor status is not obvious to third parties must identify themselves as such to avoid creating an impression in the minds of members of the public that they are Government officials. The Contractor shall ensure that all documents or reports produced by Contractors are suitably marked as Contractor products or that Contractor participation is appropriately disclosed.

1.11.1 Contractor Workspace Identification:

If applicable, Contractor workspace (staging site, on-site office, desk or work area, etc.) shall contain a sign signifying the space is occupied by Contractor employee(s) to ensure that Government employees and the public know that they are not Government employees. The proposed sign shall be submitted to the Contracting Officer via the COR for review and acceptance as requested after contract award. Workspace sign location shall be coordinated with the COR. Contractor employees shall identify themselves by name and company affiliation when answering the telephone, presenting briefings, conducting or attending meetings, and seminars. All Contractor correspondence (written, facsimile, and email display) shall include their company name. Requirements for Contractor vehicles is addressed under Part 4 paragraph 4.4.

1.11.2 Contractor Identification Badges:

Contractor personnel shall be easily identifiable through the display of badges IAW Contractor Identification (AE Reg. 27-715). For services at Child, Youth, and School Services (CYSS Brunssum) where children are present, contractor personnel shall wear assigned identification when children are present.

1.12 Contractor Travel:

RESERVED

1.13 Other Direct Costs (ODCs):

RESERVED

1.14 Data Rights:

The Government has unlimited rights to all documents and materials produced under the contract. All documents and materials, to include the source codes of software, produced under this PWS shall be Government owned and are the property of the Government with all rights and privileges of ownership and copyright belonging exclusively to the Government. These documents and materials may not be used or sold by the Contractor without written permission from the Contracting Officer. All materials supplied to the Government shall be the sole property of the Government and may not be used for any other purpose. This right does not abrogate any other Government rights.

1.15 Non-Disclosure Requirements:

Performance under this PWS may require the Contractor to access data and information proprietary to a Government agency, another Government Contractor, or of such nature that its dissemination or use other than as specified in this work statement would be adverse to the interests of the Government or others. The Contractor and Contractor personnel, shall not divulge or release data or information developed, or obtained under performance of this PWS, except to authorized Government personnel or upon written approval of the Contracting Officer. The Contractor shall not use, disclose, or reproduce proprietary data, which bears a restrictive legend, other than as specified in this PWS. All documentation showing individual names or other personal information shall be controlled and protected under the provisions of the Privacy Act of 1974, Public Law 93-579, 5 United States Code (U.S.C.) Section 552a.

1.16 Use of Government Information and Imagery:

The use of propaganda violates DoD Commercial Use of Imagery Guidelines stated at <http://www.defenseimagery.mil/products/dodimagery/commercialuse.html> The Contractor shall not cite any information (e.g., contract information, pictures, locations, etc.) obtained through this PWS on any marketing tools to include its company website.

1.17 Non-Disclosure Statements:

RESERVED

1.18 Organizational Conflict of Interest (OCI):

Contractor and subcontractor personnel performing work under this PWS may receive, have access to or participate in the development of proprietary or source selection information (e.g., cost or pricing information, budget information or analyses, specifications or work statements, etc.) or perform evaluation services which may create a current or subsequent OCI as defined in FAR Subpart 9.5. The Contractor shall notify the Contracting Officer immediately whenever it becomes aware that such access or participation may result in any actual or potential OCI and shall promptly submit a plan to the Contracting Officer to avoid or mitigate any such OCI (*Deliverable 9*). The Contractor's OCI Mitigation Plan will be determined to be acceptable solely at the discretion of the Contracting Officer. In the event the Contracting Officer unilaterally determines that any such OCI cannot be satisfactorily avoided or mitigated, the Contracting Officer may affect other remedies as he or she deems necessary, including prohibiting the Contractor from participation in subsequent contracted requirements which may be affected by the OCI.

1.19 Phase In /Phase Out Period:

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1.20 Safety:

The following provides information pertaining to Safety and Accident Prevention: the contractor is required to adhere to United States (U.S.) Army regulations, as well as Dutch and USAG Benelux-Brunssum local policies and procedures. In case of an accident the contractor shall forward one completed copy of locally required accident reports to the COR and government representative (DPW) not later than five (5) work days after the occurrence of an accident/injury to his/her personnel (*Deliverable 10*).

1.20.1 Host Nation Safety Laws and Regulations:

The Contractor shall comply with FAR 52.236-13, Accident Prevention (November 1991), the references contained therein (e.g. U.S. Army Corps of Engineers Safety and Health Requirements Manual (EM 385-1-1)), and applicable Host Nation Safety Laws and Regulations.

The requirements under FAR 52.236-13 and specifically of the EM 385-1-1 include, but are not limited to the following: (a) Before initiation of work at the job site, an Accident Prevention Plan, written by the Contractor for the specific work and hazards of the contract and implementing in detail the pertinent requirements of EM 385-1-1, will be reviewed for acceptance by designated Government personnel (*Deliverable 10*). Specific requirements for development of the Accident Prevention Plan are found in 01.A and Appendix A of EM 385-1-1. EM 385-1-1 is available online at www.usace.army.mil.

1.20.1.1 Accident Prevention and Safety Compliance:

In addition to other requirements for accident prevention specified in the contract, the Contractor shall comply with all Federal and State requirements governing safety and health protection on construction sites and other applicable occupational safety codes.

The Contractor shall indicate its plan for compliance considering conditions of the work described by the design, technical proposal, operational activities of the Government or others in the area of the work, identification of the coordinator, and implementation of all other accident prevention provisions of the contract. For the compliance with host nation laws and regulations, the Contractor shall comply with the non-exclusive list of duties described in this paragraph. If the Contractor determines that additional activities are required to fully comply, the Contractor shall ensure that compliance with the laws is complete.

1.20.1.2 Safety Notifications and Documentation:

The Contractor shall complete and provide safety notifications required by host nation authorities and will provide documented proof to the Contracting Officer via the COR, no later than five (5) days (*Deliverable 9*) that such notifications were made.

1.21 Environmental Compliance:

The following provides information on Environmental Compliance.

1.21.1 Environmental Compliance Plan and Program:

The Environmental Compliance Plan and Program focuses on reducing risk, preventing pollution, and maintaining regulatory compliance. USAG Benelux locations in the Netherlands are responsible for multimedia environmental compliance in accordance with the Final Governing Standards (FGS-NL 2020), U.S. Army Regulation (AR) 200-1 Environmental Protection and Enhancement, Army in Europe Regulation (AER) 200-1 Environmental Quality, applicable Host Nation laws and regulations, and USAG Benelux internal environmental requirements and procedures.

To support this responsibility, all service records, reports, certificates, and logbooks generated under this contract shall be provided to the COR. All documents shall be made accessible to the DPW Environmental Division for review upon request.

The Contractor shall provide all required inspection and re-inspection certificates no later than fourteen (14) days prior to their expiration date (*Deliverable 10 & PRS#8*).

The Contractor shall comply with all applicable Dutch national and municipal laws, regulations, and standards governing the handling, storage, transportation, and disposal of hazardous materials and hazardous waste. Long-term storage of flammable liquids and other hazardous materials on site is prohibited unless specifically authorized and conducted in approved storage locations. The Contractor shall ensure that all contractor personnel are properly instructed in the correct use, handling, storage, potential hazards, and required protective measures associated with hazardous materials. Third parties shall not be endangered by the use of hazardous materials. Safety Data Sheets (SDS) shall be readily available for all hazardous materials used in performance of this contract.

The Contractor shall comply with the USAG Benelux Integrated Solid Waste Management Plan (ISWMP). Copies of this plan shall be provided annually by the COR.

The Contractor shall, at its own expense, immediately contain and remediate any contamination, including soil or groundwater contamination, caused by hazardous materials or substances brought onto the Garrison or resulting from contractor activities. Any hazardous substance spill shall be reported immediately to the DPW Environmental Division.

Contractor-supplied personnel performing work under this PWS shall comply with all applicable regulations, rules, guidance, policies, permits, and laws governing U.S. operations in the Netherlands, including at a minimum:

- a) Final Governing Standards (FGS-NL) for the Netherlands (2020);
- b) Applicable Host Nation statutes, standards, permits, and environmental regulations related to environmental compliance, protection, management, and pollution prevention.

The Contractor shall provide all required job-specific certifications prior to performing tasks requiring such certification.

Environmental compliance standards and guidance are available through the USAG Benelux DPW Environmental Division. Consultation with the Environmental Office is recommended to ensure environmental compliance requirements are adequately addressed.

1.22 Required Training:

The following provides information on training requirements.

1.22.1 Anti-Terrorism (AT) Level I Training:

All Contractor employees, including subcontractor employees, who are employed under the contract, shall complete AT Level I Training within twenty (30) days (*Deliverable 12*) after contract start date or effective date of incorporation of this requirement into the contract, whichever applies, or twenty (30) days after employment of new personnel. Contractor personnel travelling overseas shall complete area of responsibility (AOR) specific AT awareness training for U.S. based Contractor employees and associated subcontractor employees. The Contractor shall submit certificates of completion (*Deliverable 12*) for each affected Contractor employee and subcontractor employee to the COR (or to the Contracting Officer, if a COR is not assigned) within twenty (30) days after completion of this training by all employees and subcontractor personnel. AT Level I awareness training is available at <https://jkodirect.jten.mil>. Contractor personnel shall complete refresher training (*Deliverable 12*) every twelve (12) months except AOR specific AT Level I training which shall be completed as required by local AT policies

If training cannot be conducted on the website, the Contractor shall coordinate with the COR who will contact the local AT Officer at their assigned location for classroom training. Contractor personnel completing training in the classroom

will receive a certificate or the training attendance roster signed by the AT Officer. Verification of the training shall be provided to the COR within twenty (30) days after completion of the training (*Deliverable 12*).

1.22.2 OPSEC Training:

All Contractor employees, including subcontractors, shall complete Level I OPSEC training within twenty (30) days of employment under this PWS (*Deliverable 13*). Verification of the training shall be provided to the COR within twenty (30) days after completion of the training (*Deliverable 13*). OPSEC Level I training is available at <http://www.cdse.edu/catalog/elearning/GS130.html>. Contractor personnel must complete refresher training every 12 months (*Deliverable 13*). Verification of the training shall be provided to the COR within twenty (30) days after completion of the training (*Deliverable 13*).

If training cannot be conducted on the website, the Contractor shall coordinate with the COR, who will contact the local OPSEC Officer at their assigned location for classroom training. Contractor personnel completing training in the classroom will receive a certificate or the training attendance roster signed by the OPSEC Officer. Verification of the training shall be provided to the COR within twenty (20) days after completion of the training (*Deliverable 14*).

1.22.3 iWATCH Training:

The Contractor and all associated subcontractors shall brief all employees on the local iWATCH program (training standards provided by the requiring activity ATO). This locally developed training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity to the COR. This training shall be completed within twenty (30) days after the contract is awarded and within twenty (30) days after new employees commence contract performance, with the results reported to the COR within twenty (30) days after the contract is awarded and within twenty (30) days after new employees commence contract performance (*Deliverable 14*). Verification of the training shall be provided to the COR within twenty (30) days after completion of the training (*Deliverable 14*). Training slides can be found on the Army OneSource website: <http://www.myarmyonesource.com/familyprogramsandservices/iwatchprogram/default.aspx>.

1.22.4 Information Assurance (IA) Training:

RESERVED

1.22.5 Personally Identifiable Information (PII):

RESERVED

1.23 Notice to Proceed:

RESERVED

1.24 Electromagnetic Compatibility Directive:

RESERVED

1.25 Government Property and Services:

RESERVED

1.26 Conventional Forces Europe (CFE) Treaty Verification Inspection:

Contractor-owned or controlled structures and containers located on U.S. Government installations are subject to short-notice inspections by CFE treaty inspection teams. The Government will endeavor to provide the Contractor as much notice as possible in the event of such inspections. The contractor shall provide access to such structures and containers at the request of the Government. The Contractor shall report such inspections to the KO through the COR in writing within ten (10) calendar days of completion of the inspection. Contractor-owned or controlled structures and/or containers on U.S. installations may have to be inspected to comply with the CFE treaty even if Contractor personnel are unavailable to provide access. Any costs borne by the Contractor or any impact on contract performance as a result of the inspection will be a matter of negotiation between the Contractor and the KO.

PART 2 DEFINITIONS & ACRONYMS

2 Definitions and Acronyms

2.1 Definitions:

Although not inclusive of every term used within this PWS, the following provides a list of definitions used throughout this PWS and commonly used in the acquisition field.

Contracting Officer – means a person with authority to enter into, administer, and or terminate contracts, and make related determinations and findings on behalf of the Government. Note: The only individual who can legally bind the Government.

Contracting Officer's Representative (COR) – As defined in DFARS 202.101, means an individual designated and authorized in writing by the Contracting Officer to perform specific technical or administrative functions. Department of Defense Instruction (DoDI) 5000.72, Part II Definitions states the following when defining a COR: “Defined in subpart 202.101 of Reference (f). Any individual delegated responsibilities pursuant to subpart 1.602-2 of Reference (e), regardless of local terminology, must be certified in accordance with this instruction. For example, local terminology can be COR, contracting officer’s technical representative, technical point of contact, technical representative, alternate COR, administrative COR, assistant COR, line item manager, task order manager, quality assurance personnel, quality assurance evaluator, or COR management.” In addition, Army Regulation 70-13, Chapter 2, paragraph 2-2g, states, in part, the following when providing support personnel to assist the COR, “...These other surveillance support personnel may serve as on-site representatives of the COR in performance of actual contract surveillance if they meet all COR requirements and have been appointed by the Contracting Officer as alternate CORs.”

Contractor – means a supplier or vendor awarded a contract to provide specific supplies or service to the Government. The term used in this PWS refers to the prime.

Contractor-acquired property - means property acquired, fabricated, or otherwise provided by the Contractor for performing a contract, and to which the Government has title.

Day – means, unless otherwise specified, a calendar day.

Defective Service – means service output that does not meet the standard of performance associated with the Performance Work Statement.

Deliverable – means anything that can be physically delivered, but may include non-manufactured things such as meeting minutes or reports.

Government-furnished property – means property furnished to the contractor the Government and subsequently furnished to the Contractor for performance of a contract. Government-furnished property includes, but is not limited to, spares and property furnished for repair, maintenance, overhaul, or modification. Government-furnished property also includes Contractor-acquired property if the Contractor-acquired property is a deliverable under a cost contract when accepted by the Government for continued use under the contract.

Government property – means all property owned or leased by the Government. Government property includes both Government-furnished property and Contractor-acquired property. Government property includes material, equipment, special tooling, special test equipment, and real property. Government property does not include intellectual property and software.

Preventive Maintenance (PM) – the systematic inspection, servicing and care of equipment, utility systems, buildings, structures and grounds facilities for the purpose of detecting and correcting incipient failures and accomplishing routine

maintenance tasks. Scheduled and correctly performed PM is recognized as an effective means of managing costs of facilities operations by reducing unexpected system failures and emergency repair costs. It contributes to sustainment of mission capabilities and reliability by enabling the scheduling of maintenance downtime in coordination with operational mission requirements. Scheduled PM maintains energy efficiency of installed systems.

Property Administrator – means an authorized representative of the Contracting Officer appointed in accordance with agency procedures, responsible for administering the contract requirements and obligations relating to Government property in the possession of a Contractor.

High Level Objective (HLO) – means a key overarching result-based objective for a project necessary to achieve the project’s vision. HLOs are similar to Level 2 in a Work Breakdown Structure. Each HLO may contain several statements to flesh out the areas necessary to meet the objective.

Physical Security – means that part of security concerned with physical measures designed to safeguard personnel; to prevent unauthorized access to equipment, installations, material, and documents; and to safeguard against espionage, sabotage, damage, and theft.

Quality Assurance – (or Government contract quality assurance) means the various functions, including, inspection, performed by the Government to determine whether a Contractor has fulfilled the contract obligations pertaining to quality and quantity.

Quality Assurance Surveillance Plan (QASP) – means the key Government-developed surveillance process document and is applied to Performance-Based Service Contracting (PBSC). The QASP is used for managing Contractor performance assessment by ensuring that systematic quality assurance methods validate that Contractor quality control efforts are timely, effective, and are delivering the results specified in the contract or task order. The QASP directly corresponds to the performance objectives and standards (i.e., quality, quantity, timeliness) specified in the Performance Work Statement (PWS). It provides specific details on how the Government will survey, observe, test, sample, evaluate, and document Contractor performance results to determine if the Contractor has met the required standards for each objective in the PWS. The QASP, with very few if any exceptions, is an internal to Government document.

Quality Control – means all necessary measures taken by the Contractor to assure that the quality of an end product or service shall meet contract requirements.

Statement – means the specific results-based activities required to satisfy HLOs. A statement contains a result, the context of the statement, and the required action(s). Statements focus on “what” is to be accomplished; however they are not prescriptive in describing “how” the outcome is to be achieved. Each HLO may have several statements to flesh out the areas necessary to meet the objective. Statements are similar to Level 3 in a Work Breakdown Structure.

Subcontractor – means one that enters into a contract with a prime Contractor. The Government does not have privity of contract with the subcontractor.

Unscheduled Maintenance – non-recurring maintenance or unscheduled repair work required to preserve and sustain a real property facility or its components due to unforeseen occurrences. The work is corrective in nature and not generally used for recurring tasks. Unscheduled maintenance is performed through the issuance of DMOs (less than the 32 work-hours threshold) and PWOs (above the 32 work-hours threshold). The purpose of a DMO is to provide the Government with a means of remedying unplanned/unscheduled work.

Work Day – the number of hours per day the Contractor provides services in accordance with the contract.

Work Week – Monday through Friday, unless otherwise specified.

2.2 Acronyms:

Although not inclusive of every term used within this PWS, the following provides a list of acronyms used throughout this PWS or commonly used in the acquisition field.

ACOR	Alternate Contracting Officer's Representative
AE	Army in Europe
AFARS	Army Federal Acquisition Regulation Supplement
AHA	Activity Hazard Analysis
AOR	Area of Responsibility
AR	Army Regulation
AT	Anti-terrorism
ATCTS	Army Training Certification Tracking System
CAC	Common Access Card
CCE	Contracting Center of Excellence
CFE	Conventional Forces Europe
CFR	Code of Federal Regulations
CONUS	Continental United States (excludes Alaska and Hawaii)
COR	Contracting Officer's Representative
COTS	Commercial-Off-the-Shelf
DA	Department of the Army
DD250	Department of Defense Form 250 (Receiving Report)
DD254	Department of Defense Contract Security Requirement List
DFARS	Defense Federal Acquisition Regulation Supplement
DMDC	Defense Manpower Data Center
DoD	Department of Defense
FAR	Federal Acquisition Regulation
GFP	Government Furnished Property
HLO	High Level Objective
HIPAA	Health Insurance Portability and Accountability Act of 1996
IA	Information Awareness
IAC	Installation Access Control
IACO	Installation Access Control Office
ID	Identification
IGCE	Independent Government Cost Estimate
IT	Information Technology
JTR	Joint Travel Regulation
JPAS	Joint Personnel Adjudication System
KO	Contracting Officer
OCI	Organizational Conflict of Interest
OCONUS	Outside Continental United States (includes Alaska and Hawaii)
ODC	Other Direct Costs
OPSEC	Operations Security
PA	Property Administrator
PII	Personally Identifiable Information
PIPO	Phase In/Phase Out
PMP	Preventive Maintenance Plan
POC	Point of Contact
PRS	Performance Requirements Summary
PWS	Performance Work Statement
QA	Quality Assurance
QAP	Quality Assurance Program

QASP	Quality Assurance Surveillance Plan
QC	Quality Control
QCP	Quality Control Program
RA	Requiring Activity
RCO	Regional Contracting Office
SCIOS	Stichting Certificering Onderhoud en Inspectie van Stookinstallaties (Foundation Certification Maintenance and Inspection of Combustion Installations)
SIGE	Safety and Health Protection Plan
TE	Technical Exhibit
USAREUR	U.S. Army Europe
UOM	Unit of Measure

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PART 3 GOVERNMENT PROPERTY (GP) AND SERVICES

3 Government Property and Services

3.1 Services:

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3.2 Facilities:

The Government will provide the contract employee access to all required mechanical, electrical and technical rooms to support daily tasks in performance of this PWS. The designated space shall be identified to the contractor during the post award conference or prior to the contract performance start date.

3.3 Utilities:

The Government will provide all utilities in the facility to be available for the Contractor's use in performance of the tasks outlined in this PWS. The Contractor will instruct employees in utilities conservation practices. The Contractor will operate under conditions that preclude the waste of utilities, which include turning off the water faucets or valves, light switches, etc. after using the required amount to accomplish tasks requiring the use of the utilities.

3.4 Equipment:

The Government will provide Building Automated System hardware and software for use in providing maintenance and adjusting to the HVAC systems. These systems will be available on-site for day-to-day use but will not be assigned nor checked-out to the contractor. Any repairs discovered during maintenance of the equipment may be ordered upon modification of the contract.

3.5 Special Tooling and Test Equipment:

RESERVED

3.6 Materials:

RESERVED

3.7 Training:

The Government will provide the following training: AT Level 1 Awareness, OPSEC, iWATCH.

3.8 Common Access Cards (CAC):

RESERVED

PART 4 CONTRACTOR FURNISHED ITEMS AND SERVICES

4 Contractor Furnished Property and Services

4.1 General:

The Contractor shall furnish all services, personnel, equipment, supplies, transportation, vehicles, tools, materials, supervision, and other items necessary to perform services as described in the PWS and associated contract documents, except for any items specified in Part 3, Government Property, Equipment and Services.

4.1.1 Proprietary Equipment Service Requirements:

The Contractor must be able to provide services on the following proprietary equipment listed in section 4.1.1 (1-6) below. The Contractor shall provide these services within their company or by sub-contracting the requirements to a company that can work on these systems by providing the COR a Memorandum of Agreement, Memorandum of Understanding, and/or written statement that this function can be fulfilled (*Deliverable 15 and PRS#1*).

1. Air Treatment: System Air
2. Heat Pumps: Carrier
3. Control Boxes: Priva Blue ID
4. Airco: Daikin
5. Boiler: Viessmann
6. Dehumidifier: Industrial Air

4.1.2 SCIOS Certification:

The Contractor or subcontractor performing inspections, maintenance, or testing on combustion installations shall be SCIOS-certified and shall use SCIOS-certified technicians.

Applicable SCIOS scopes include, but are not limited to:

- Scope 1 / 2 (EBI and PO for combustion systems)
- Scope 6 (Periodic Inspection – PI)
- Scope 7a / 7b (inspection of gas and fuel distribution piping)
- Scope 7c (emission and efficiency measurements, where required by environmental permits)
- Scope 14 (geothermal ground energy systems), where applicable

4.2 Secret Facility Clearance:

RESERVED

4.3 Materials:

The Contractor will provide necessary Personal Protective Equipment (PPE) for the performance of the contract.

4.4 Vehicles and Equipment:

The Contractor shall furnish all necessary vehicles and equipment to perform the work described in this PWS.

4.5 Background Checks:

The contractor must adhere to the requirements outlined in paragraph 1.6.1 of this PWS.

PART 5 SPECIFIC TASKS

5 Specific parts

5.1 Basic Services:

The Contractor shall provide preventive maintenance (PM), inventory, and labeling services for Heating, Ventilation, Air Conditioning, and Combination (HVAC/Luchtbehandelingskast–LBK) systems at USAG Benelux – Brunssum, CYSS, and APS Egelshoven.

Systems and components covered under this requirement include, but are not limited to: indirect units, direct units, split units, gas-fired boilers, electric boilers used for space heating, furnaces, compressors, control systems, burner assemblies, expansion vessels, drums, temperature controls, air curtains, air filters, condenser units, exhaust fans, ventilation fans, and recirculation units.

All preventive maintenance shall be performed in accordance with manufacturer specifications, Host Nation laws, and applicable Army regulations referenced in Part 6 of this PWS.

5.2 Heating, Ventilation, and Air Conditioning Preventative Maintenance, Inventory and Labeling:

Services under this requirement are recurring and divided into three sub-tasks: (1) Preventive Maintenance, (2) Inventory, and (3) Labeling.

The Government will issue Delivery Orders (DOs) under this contract as services are required. Upon receipt of a DO, the Contractor shall coordinate with the COR to schedule specific services.

All services performed under this requirement shall be documented in maintenance logbooks within one (1) business day of completion (PRS #2). All equipment certifications executed under this contract shall be provided to the COR within five (5) business days of issuance. (PRS #3).

The Contractor is responsible for obtaining all manufacturer documents, manuals, and specifications necessary to perform PM. Deviations from manufacturer guidance due to updated laws or standards require prior written COR approval.

5.2.1 Recurring Services:

This section defines the maintenance requirements for all Heating, Ventilation, and Air-Conditioning (HVAC) systems and components covered under this contract. The Contractor shall maintain all HVAC equipment identified in Technical Exhibit C (TE-C), including but not limited to heating and combustion systems, hydronic heating and cooling systems, ventilation and air-handling systems, refrigeration and air-conditioning systems, control-air compressors, dehumidifiers, and all associated auxiliary equipment.

Maintenance shall be performed in accordance with manufacturer recommendations, applicable Host Nation regulations, U.S. Army policies, environmental permits, SCIOS requirements, and industry best practices. All HVAC systems shall be maintained to ensure safe, reliable, and energy-efficient operation and to support mission readiness at all times.

The Contractor shall perform both preventive maintenance and demand maintenance (PM and DMO) in accordance with this PWS, applicable Performance Requirements Summary (PRS) items, and any direction provided by the COR. All system components included in TE-C, regardless of installation date, manufacturer, or capacity, fall within the scope of this section unless specifically excluded in paragraph 1.4 of this PWS.

5.2.1.1 Preventative Maintenance (General):

The Contractor shall perform Preventive Maintenance (PM) on all systems and equipment servicing the installations identified in paragraph 5.1 and as listed in Technical Exhibit C (TE-C). If systems or equipment are added, removed, or replaced during the contract period, TE-C and associated pricing shall be updated through formal contract modification. Preventive Maintenance shall ensure that all systems:

- Function as designed;

- Operate to optimize component life;
- Minimize service interruptions;
- Reduce environmental impact;
- Maximize energy efficiency.

Any malfunction identified during PM shall be recorded and reported to the COR within four (4) hours of discovery (*Deliverable 3*).

5.2.1.1.1 Heating and Combustion Systems:

The Contractor shall perform annual PM and functional inspection of all gas-fired and electrically heated space-heating systems identified in this PWS. All work shall comply with manufacturer requirements, applicable environmental permits, and the Dutch Besluit Activiteiten Leefomgeving (BAL).

Work on combustion systems shall be performed in accordance with the applicable provisions of the BAL governing environmentally harmful activities related to combustion installations, including but not limited to:

- BAL Chapter 3 – Environmentally harmful activity: combustion installation (3.2.1)
- BAL Chapter 4 – Large combustion installations (4.3)
- BAL Chapter 4 – Small and medium combustion installations using standard fuels (4.126)
- BAL Chapter 4 – Medium combustion installations using non-standard fuels (4.127)

The Contractor shall ensure that all inspections, emission measurements, and maintenance activities are performed in accordance with the applicable BAL requirements for the specific installation category and capacity.

For gas-fired installations, inspections and adjustments shall be performed by a SCIOS-certified company in full compliance with the SCIOS certification scheme. Applicable SCIOS Scopes typically include:

- **Scope 1 or #** (combustion systems),
- **Scope 6** (hydraulic components),
- **Scope 7c** (efficiency and emission testing, where required).

PM tasks shall include, but are not limited to:

- Combustion analysis and burner adjustment;
- Verification of temperature and pressure limits;
- Testing of safety controls, flame supervision, and gas-tightness;
- Inspection of combustion air supply and flue-gas exhaust;
- Verification of condensate drainage systems;
- Submission of SCIOS certificates and test reports to the COR.

Any deviation or unsafe condition shall be reported immediately as a DMO.

5.2.1.1.2 Hydronic Heating and Cooling Systems:

The Contractor shall perform annual PM and inspection of all closed-loop hydronic systems serving heating or cooling functions. This includes circulation pumps, expansion vessels, buffer vessels, automatic filling units, dirt and air separators, control valves, sensors, and all associated piping.

PM tasks shall include, but are not limited to:

- Inspection for leaks, corrosion, vibration, abnormal noise, and structural support;
- Verification of circulation pump operation, setpoints, and direction of flow;
- Testing of isolation, balancing, and drain valves;
- Verification of expansion vessel pre-charge pressure and diaphragm integrity;
- Inspection of buffer vessels, insulation, and instrumentation;

- Flushing or draining to remove sediment, sludge, or magnetite where indicated;
- Verification of automatic air vents and filling units;
- Recording all readings, measurements, and maintenance actions in the HVAC logbook.

Any hydronic component showing signs of failure, waterlogging, pressure imbalance, or performance degradation shall be corrected or reported immediately as a DMO.

Backflow prevention devices (Type CA or Type BA) that form part of, or are directly connected to, the potable water installation fall under the scope of the Waterlines contract. The Contractor shall not perform testing, inspection, certification, adjustment, or maintenance of these potable-water components under this HVAC contract. Any deficiencies observed in such backflow prevention devices shall be reported immediately to the COR.

5.2.1.1.3 Air-Side Systems (Ventilation and HVAC Units):

The Contractor shall perform annual PM and certification for all air-side HVAC systems, including ventilation units, exhaust fans, air-handling units (LBKs), packaged units, split systems, duct-free units, and all other air-side components. PM tasks shall include:

- Cleaning, lubrication, and disinfection of fans, motors, coils, filters, drains, dampers, and casings;
- Inspection of ductwork, air intakes, filter frames, bearings, and electrical components;
- Verification of belt condition and fan balance;
- Replacement of filters (at least twice annually) and belts (at least once annually);
- Measurement of indoor climate parameters where applicable;
- Recording all test results and adjustments.

All waste and removed materials shall be disposed of in accordance with applicable environmental regulations.

5.2.1.1.4 Dehumidifiers (Stand-Alone Units):

The Contractor shall perform annual PM and inspection of all stand-alone dehumidifiers serving warehouses or storage areas.

PM tasks shall include:

- Inspection for leaks, corrosion, or mechanical damage;
- Cleaning or replacement of filters and heat exchangers;
- Inspection and testing of refrigeration circuits and refrigerant charge;
- Calibration and testing of humidity and temperature controls;
- Verification of electrical safety, wiring, and grounding.
- Quarterly replacement of air filters on all dehumidifier units, in addition to the annual preventive maintenance requirements.

All findings shall be recorded and deviations reported as DMOs.

5.2.1.1.5 Refrigeration and Air-Conditioning Systems:

The Contractor shall perform PM and mandatory leak testing on all refrigeration and air-conditioning systems in accordance with FGS-NL Chapter 2 (Air Emissions) and EU F-Gas Regulation (EU 517/2014), and manufacturer requirements.

Leak testing frequency shall be:

- Systems < 5,000 kg CO₂-equivalent → annual leak test (PRS #5);
- Systems ≥ 5,000 kg CO₂-equivalent → semi-annual leak test (PRS #6).

Systems equipped with an automatic leak detection system shall be inspected in accordance with the inspection intervals defined in the applicable EU F-Gas Regulation (currently Regulation (EU) 2024/573, Article 7), including any required increase in inspection frequency based on the CO₂-equivalent charge thresholds.

PM tasks shall include:

- Cleaning and inspection of compressors, evaporators, condensers, coils, piping, and valves;
- Verification of refrigerant charge, pressures, leak integrity, and superheat/subcool readings;
- Functional testing of sensors, thermostats, electronic expansion valves, controllers, and safety devices;
- Inspection of mounting hardware, vibration isolators, electrical connections, and cabinet integrity;
- Documentation of refrigerant use, recovery, disposal, and leak-test results, and corrective action in accordance with EU F-Gas rules.

Following any leakage repair on an F-gas refrigeration or air-conditioning system, the Contractor shall perform a mandatory follow-up verification leak test at a later date, in accordance with Article 4 of Regulation (EU) 2024/573. Both the repair and the verification test shall be recorded in the system's F-gas logbook.

For each refrigeration or air-conditioning system containing F-gases, the Contractor shall maintain a system-specific F-gas logbook. At a minimum, the logbook shall include:

- system identification (equipment ID, location, refrigerant type and total charge);
- company and technician F-gas certifications (BRL 100 and BRL 200);
- all leak inspections and leak-test results, including the method used to perform the leak test;
- all leakage repairs and the associated mandatory follow-up verification leak tests;
- all refrigerant charging and recovery activities, including quantities and certification of recovered refrigerant;
- Identification of the technician performing the leak test, including name and date of performance.

Where applicable under PED Category 2, the logbook shall also reference the pressure test certificate, the conformity declaration, and the refrigeration schematic diagram. The Contractor shall maintain these records for the duration of the contract and make them available to the COR upon request.

Refrigerant, filters, and consumables are included in the contract price.

5.2.1.1.6 Control Air and Compressor Systems:

The Contractor shall perform annual PM and inspection of all air compressors and auxiliary equipment used to supply HVAC control air or pneumatic actuators.

PM tasks shall include:

- Inspection of compressor units, dryers, filters, receivers, and piping for leaks or vibration;
- Checking oil level (if applicable), belts, and motor operation;
- Draining condensate and testing automatic drains;
- Verification of air pressure, dryer operation, and safety relief devices;
- Recording all readings and maintenance actions in the HVAC logbook.

Compressors serving workshop or general facility use are excluded.

5.2.1.1.7 Fuel and Gas Distribution Systems:

The Contractor shall perform PM, inspection, and SCIOS-certified testing of all gas and fuel distribution piping serving HVAC combustion systems, up to the appliance connection point.

All work shall comply with:

- SCIOS Scope 7a / 7b;
- NEN 1078 and NEN 3028;
- Applicable environmental permits.

All inspection, testing, and maintenance activities on fuel and gas distribution piping serving HVAC combustion systems, up to the appliance connection point, shall be performed by personnel certified for SCIOS Scope 7b. All work shall

comply with the SCIOS Certification Scheme, the Dutch Activiteitenbesluit, NEN 1078, and NEN 3028. The Contractor shall submit all SCIOS 7a/7b/7c reports, test results, and certificates to the COR and maintain these records in the HVAC maintenance logbook.

SCIOS reports shall be submitted to the COR and shall be retained in the HVAC maintenance logbook for the full duration of the contract.

5.2.1.1.8 Documentation Requirements:

For each PM activity, the Contractor shall document:

- Equipment ID and location;
- Date and nature of inspection;
- Measurements, observations, and test results;
- Deficiencies and corrective actions;
- Name and qualification of personnel.

The Contractor shall provide records to the COR upon request and include all PM activities in the quarterly maintenance summary report.

5.2.2 Inventory and Labeling:

The contractor shall be responsible for creating an inventory of all systems on each installation. This inventory shall be updated as new systems are installed, or as others are removed. The initial or modified inventory shall be provided to the COR within 30 days of the issuance of a delivery order requesting the services (*Deliverable 16*). During the creation of the inventory for each installations, the contractor shall provide and permanently affix pre-numbered "Identification Tags" to all equipment. The numbering sequence provided on each identification tag shall be created as directed by the COR. This sequence will be comprised of numbers which identify the installation, building, room number, location, and type of each system identified.

5.2.3 PM Documentation:

If the Contractor detects a deficiency or malfunction of the equipment during PM and repairs are required, the Contractor shall document the finding on the PM work certificate and notify the COR immediately. The COR will determine whether a Demand Maintenance Order (DMO) will be initiated. (*Deliverable 18*)

At a minimum, the work certificate shall contain the following information:

- A. Date, time, and printed and signed name of the individual who performed the PM work on item (Equipment Number, Name, and Location according to Inventory List).
- B. A list of all materials and spare parts used for the required repairs.
- C. Cost estimates for any required materials and spare parts for repairs.

5.3 Demand Maintenance Order (DMO):

Non-recurring maintenance or unscheduled repair. Work required to preserve and sustain a real property facility or its components due to unforeseen occurrences. The work is corrective in nature and not generally used for recurring tasks. Unscheduled maintenance is performed through the issuance of DMO. The purpose of a DMO is to provide the Government with a means of remedying unplanned / unscheduled work occurring outside of the normal recurring work. The Contractor and Government shall utilize DMOs to accomplish all unplanned / unscheduled work at the discretion of the COR. The Government shall combine multiple repair requirements received for the same trade in the same building or structure at the same time into one DMO, as long as the hourly threshold is not exceeded. A DMO may involve multiple crafts such as both electrical and carpentry work and more than one related task at a common work site.

Upon receipt of a DMO, the Contractor shall provide an informal estimate of labor work-hours and materials cost of the repair to the COR prior to starting work. The COR/KO shall give instructions to the Contractor to proceed if the DMO is approved. The Contractor shall only proceed with the work once approval is given by the COR/KO. Approved DMO work must stop once the work is expected to exceed 32 hours labor hours, the contractor shall then notify the COR/KO and not proceed until further guidance is given.

5.3.1 DMO Performance Requirements.

There are 3 priorities for DMOs. They are classified in Priority 1, 2 and 3. The Government will determine DMO types and priorities according to set criteria, and reserves the right to reclassify DMO priorities as it deems necessary. When notified by the COR, the contractor shall respond to DMOs and complete maintenance repairs in accordance with the categorized DMO:

CATEGORY	DEFINITION	EXAMPLE	TIMELINE REQUIREMENT
Priority 3	For work that does not meet the criteria of emergency/critical or urgent priority. They are utilized to correct situations that are productivity inhibitors that, if left uncorrected, will cause measurable discomfort or inconvenience to the customer or waste resources or create the need for additional minor repairs.	Examples include, but are not limited to: dripping faucets, cracked window glass, repairs to door and window hardware and screening, replacement of broken or missing tiles, minor roof leaks, minor repairs to mechanical equipment, air conditioning, etc.	These DMOs are to be completed within thirty (30) calendar days of receipt.
Priority 2	Failure in service does not immediately endanger personnel or property, but would soon inconvenience or affect the security, health, or well-being of personnel.	Examples include, but are not limited to: unstopping a sink drain where there are backup sinks, replacing overhead office lighting that significantly inhibits productivity, and unscheduled fuel deliveries, etc.	These DMOs are to be completed within seven (7) calendar days from receipt, unless reduced to a Priority 3.
Priority 1	Immediate action is required to eliminate life threatening or serious injury hazards to personnel	Examples include, but are not limited to: Gas, oil, and steam leaks, water or sewage line breaks, Building floods, Safe and Secure Alarms/Doors, Sewage backing up into building, Total power outage in a building, Downed high voltage power line, Loss of heat during periods of cold weather, Locksmithing, damages from Natural Disasters.	Responded to as soon as possible and NLT one (1) hour after the request. Emergency/Critical DMOs are completed within twenty-four (24) hours of the request, unless the urgency of the work has been reduced to a Priority 2 or 3.

5.3.2 Notification of DMO Timeline Issues:

The contractor shall notify the COR immediately if the DMO Timeline Requirement will not be met.

5.3.3 Equipment Replacement Parts for DMOs (unscheduled tasks):

Contractor shall furnish all parts and materials required for repair of equipment under this contract and shall complete services during the initial scheduled maintenance or repair call when parts are available. Prior to replacing any part the Contractor, provide a full estimate of repair/replacement for consideration prior beginning any work.

Individual parts costing over € 2,500 to repair/replace must be completed by obtaining at least three (3) quotes. These quotes will be provided to the Contracting Officer via the COR, commencement of work is subject to approval by the Contracting Officer prior to beginning work. (*Deliverable #19*).

5.3.4 Maintenance /Repair Reports and Invoicing:

Contractor shall manage all assigned projects from initial receipt until project completion. The Contractor shall ensure all projects are completed IAW the PWS and Contractor's approved schedule. The Contractor shall keep records and prepare reports for all services performed under this contract.

5.3.5 Invoicing Requirements:

The Contractor shall submit all invoices (electronically via WAWF) within 30 days from completion of work (*Deliverable #20*). At minimum all invoices will contain the following information:

1. Contract Number/Task Order.
2. Call/Job Number.
3. Type of system, Manufacturing Number, Inventory Number, Serial Number.
4. Location.
5. Description of Malfunction.
6. Date, start time, and completion time of services.
7. Name(s) of individual(s) performing the services.
8. Type and amount of required material and replacement parts.
9. Information or recommendations pertaining to the condition or function of the system.
10. Amount of labor hours (signed time sheets).
11. Detailed receipts for expenses and/or invoices from subcontractor reflecting above requirements (As Applicable)

5.3.6 Reporting Upon Request:

When requested by the COR or KO, the contractor shall prepare and provide reports to include but not limited to: Parts Consumption, Labor Hours, Invoice Costs, Waste Removal, Technical Data, Maintenance Schedules etc. When requested reports will be provided to the COR/KO within five (5) days. (*Deliverable #21*).

5.4 Service Contract Reporting:

5.4.1 Meetings, Conferences, and Briefings:

The Contractor shall attend, participate in, and furnish input to scheduled and unscheduled meetings, conferences, and briefings that relate to the functions and services herein as required by the Government to provide effective communication and impart necessary information. The Project Manager or designated representative shall attend meetings as requested by the Government. Meeting attendees shall at times include Contractor managerial, supervisory, and other personnel knowledgeable of the subject matter. Meetings may start or end outside of regular duty hours.

5.4.2 System for Award Management (SAM) Service Contract Report (SCR, (if required)):

The Contractor shall report ALL Contractor labor hours (including subcontractor labor hours) required for performance of services provided under this contract for the *{Insert Component's Name}* via a secure data collection site. The Contractor is required to completely fill in all required data fields through the following web address: www.SAM.gov. Reporting inputs will be for the labor executed during the period of performance during each Government FY, which runs from October 1 through September 30. While inputs may be reported any time during the FY, all data shall be reported no later than October 31 of each calendar year. Contractors may direct questions to the help desk by clicking "View Assistance for SAM.gov" which is located at the top of the SAM.gov website. From there, you can select "Contact Our Service Desk" which will allow you to contact SAM.gov directly. If contract period of performance ends prior to September 30, the Contractor has 30 calendar days from end date of the contract to complete the SAM SCR requirement (*Deliverable #17*).

Steps for Submitting a Service Contract Report (SCR)

1. Go to www.sam.gov and log in.
2. Select Entity Registrations and then select Service Contract Reporting.
3. SAM displays your entities which have service contracts and meet the reporting criteria. Select View by entity to see the service contracts for each entity.
4. Next, select Add for the service contract against which you want to create a Service Contract Report. Each service contract which meets the FAR Subpart 4.1703 reporting thresholds is displayed.
5. You will be taken to the Complete Service Contract Report page. SAM displays the contract details and allows you to report. You are required to enter the following information:
 - a. **Total Amount Invoiced:** Total dollar amount invoiced for services performed during the previous Government fiscal year under the contract (this amount should include the prime and any subcontract amount).
 - b. **Prime Contractor Hours Expended:** Prime contractor direct labor hours expended on the services performed during the previous Government fiscal year. The amount you enter is automatically divided by 2,080 hours to calculate a Full Time Employee (FTE) equivalent, displayed under the Prime Contractor Hours Expended as Prime Contractor FTEs.
6. Report any required Tier 1 subcontractor information by selecting the Add Tier 1 Subcontract Information button.
7. When you are ready to submit the report, select Submit. This saves your report and returns you to the Select Service Contract page where you can create other SCRs or edit an existing SCR.

Steps for Editing a Submitted Report

1. Follow steps 1-4 above. Any previously submitted SCR will have a View/Edit button instead of an Add button.
2. Select View/Edit. You will be able to view the current SCR, edit available fields, and resubmit the record. You can also delete previously entered information altogether.

PART 6 APPLICABLE PUBLICATIONS

6 Applicable Publications (Current Editions):

The following publications, manuals, regulations, etc. are mentioned in this PWS and are listed below.

- 6.1.1 Army in Europe (AE) Regulation 190-16; Installation Access Control
- 6.1.2 Army in Europe (AE) Regulation 604-1
- 6.1.3 The Army Physical Security Program (AR 190-13)
- 6.1.4 FAR Subpart 42.5
- 6.1.5 BRL 100 – Certification scheme for companies working on stationary refrigeration, air-conditioning, and heat-pump equipment containing fluorinated greenhouse gases (F-gases).
- 6.1.6 BRL 200 – Certification scheme for personnel working on stationary refrigeration, air-conditioning, and heat-pump equipment containing fluorinated greenhouse gases (F-gases).
- 6.1.7 Veiligheid, gezondheid en milieu Checklist Aannemers (VCA) certificate
- 6.1.8 Contractor Identification (AE Reg. 27-715)
- 6.1.9 Privacy Act of 1974, Public Law 93-579, 5 United States Code (U.S.C.) Section 552a
- 6.1.10 Organizational Conflict of Interest (OCI), FAR Subpart 9.5
- 6.1.11 Accident Prevention (November 1991), FAR 52.236-13
- 6.1.12 Accident Prevention, EM 385-1-1
- 6.1.13 Safety and Health Protection Plan (SIGE), BaustellIV, dated 10 June 1998
- 6.1.14 Environmental Policy at the bottom of the U.S. Army garrison Benelux homepage (ISO14001:2004 Standards, Annex A2)
- 6.1.15 AT Level I awareness training, <https://jkodirect.iten.mil>
- 6.1.16 OPSEC Level I training is available at <http://www.cdse.edu/catalog/elearning/GS130.html>
- 6.1.17 iWATCH, <http://www.myarmyonesource.com/familyprogramsandservices/iwatchprogram/default.aspx>.
- 6.1.18 COR Definition, DFARS 202.101
- 6.1.19 COR Definition, Department of Defense Instruction (DoDI) 5000.72, Part II
- 6.1.20 COR Definition, Army Regulation 70-13, Chapter 2, paragraph 2-2g
- 6.1.21 Preventive Maintenance Plan (PMP) for facilities, property, installed equipment, infrastructure and utilities IAW AR 420-1
- 6.1.22 U.S. Army Corps of Engineers, Technical Manual (TM) 5-610, Preventive Maintenance (PM), Facilities Engineering Buildings and Structures (1979)
- 6.1.23 Guidance on operating procedures, organization and frequency of preventive maintenance services, TM 5-600 series
- 6.1.24 Technical Manual (TM) 5-810-5, Plumbing
- 6.1.25 Water Supply Systems, UFC 3-230-02 section 3.5.2
- 6.1.26 Dutch Clean Water Act of 2011 (Waterwet)
- 6.1.27 Dutch General requirements for water supply systems, NEN 1006, 15th Edition, June 2018
- 6.1.28 Transport Solutions, FGS-NL
- 6.1.29 Inspection and Maintenance of Hose Systems/Reels, NEN –EN 671-3
- 6.1.30 Testing and Calibration Laboratories, ISO 17025
- 6.1.31 FGS-NL Chapter C3.3.1.7.3
- 6.1.32 NPR 7601 – Application of carbon dioxide as a refrigerant in refrigeration systems and heat pumps
- 6.1.33 NEN 3028 – Requirements for fuel combustion installations (2022)

PART 7 ATTACHMENT AND TECHNICAL EXHIBIT LISTING

7 Attachment and Technical Exhibit List

7.1 Technical Exhibit A

Performance Requirements Summary

7.2 Technical Exhibit B

Deliverables Schedule

7.3 Technical Exhibit C

HVAC Equipment Inventory

7.4 Attachment 1

Estimated Workload Data

7.5 Attachment 2

Historical Workload Data – Not Applicable

7.6 Attachment 3

(If applicable) Government Property (GP) (The Requiring Activity shall consult with the Property Administrator when providing GP.)

7.7 Attachment 4

Signature and Approval (Internal use only)

TECHNICAL EXHIBIT A
Performance Requirements Summary (PRS)

PERFORMANCE OBJECTIVE	STANDARD	PERFORMANCE THRESHOLD ACCEPTABLE QUALITY LEVELS (AQLs)	METHOD OF SURVEILLANCE
PRS#1 - PWS para (4.1.1 and 2) The Contractor must be able to provide services on the following proprietary equipment listed in section 4.1.1 (1-6)	Throughout Delivery Order period	100%	100% Inspection
PRS#2 - PWS para (5.2.) All services performed under this requirement shall be documented in maintenance logbooks	Within 1-day of the service being performed	100%	Periodic Inspection
PRS#3 - PWS para (5.2.) All equipment certifications executed under this contract shall be provided to the COR	Within five (5) business days of issuance of the certification	100%	Random Sampling
PRS#4 - PWS para (5.2.1.1.) The contractor shall be responsible for the preventative maintenance of all systems servicing the installations identified in paragraph 5.1	Per manufacturer's recommendations	100%	Periodic Inspection
PRS#5 – PWS para (5.2.1.1.3.) The contractor shall perform ANNUAL preventative maintenance and leak tests for air conditioning systems with a CO2 equivalent < 5,000 kg	Annual	100%	Periodic Inspection
PRS#6 – PWS para (5.2.1.1.4) The contractor shall perform preventative maintenance and leak tests for air conditioning systems with a CO2 equivalent > 5,000 kg	Semi-annual	100%	Periodic Inspection
PRS#7 – PWS para (5.2.1.1.5) Annual Test/Certification HVAC-Combo Systems	Annual	100%	Random Sampling
PRS#8 – PWS para (5.2) The contractor shall ensure that all SCIOS-required inspections and certifications are valid for all applicable installations	All required inspections shall be completed no later than fourteen (14) calendar days prior to certificate expiration	100%	Periodic Inspection

**TECHNICAL EXHIBIT B
DELIVERABLES SCHEDULE**

<u>Deliverable</u>	<u>Frequency</u>	<u># of Copies</u>	<u>Medium/Format</u>	<u>Submit To</u>
<i>Deliverable 1, Quality Control Plan, 1.5.1</i>	Within 30 days after contract award and no later than 30 days prior to any anticipated changes.	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 2, Analysis of Quality Control, 1.5.1.1</i>	Within thirty 30 days upon completion of each option period or Task Order	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 3, Report issues or potential issues, 1.5.1.2</i>	Within four (4) hours of identifying those issues. Verbal report shall be followed up by written report within one (3) days	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 4, Work Plan, 1.5.4.4</i>	Ten (10) days before execution of work	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 5, Installation Access Pass Log, 1.6.3.1</i>	Within three (3) days after award and five (5) days after any changes	N/A	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 6, Installation Access Pass Turn-in, 1.6.3.2</i>	Within three (3) days	N/A	Hand Deliver	COR/KO
<i>Deliverable 7, Key Personnel Curriculum Vitae, 1.10</i>	No later than Post Award meeting	N/A	Hand Deliver/Verbal and memorandum	COR/KO
<i>Deliverable 8, Certifications, 1.10.c</i>	Within five (5) days after award and change of employee	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 9, Safety Report, 1.20.1.2</i>	Within five (5) days after each incident	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 10, Environmental compliance</i>	As required; upon issuance or renewal of certificates and reports; and no later	One each	Electronic Document via email using	COR (Access shall be provided to the

<u>Deliverable</u>	<u>Frequency</u>	<u># of Copies</u>	<u>Medium/Format</u>	<u>Submit To</u>
documentation required to demonstrate compliance with applicable environmental laws, regulations, Final Governing Standards (FGS-NL), and USAG Benelux environmental requirements, 1.21.1	than fourteen (14) calendar days prior to certificate expiration.		Microsoft Office or PDF	DPW Environmental Division upon request.)
<i>Deliverable 11</i> , Accident Prevention Plan, 1.21.1	Before initiation of work at job-site	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 12</i> , At Level 1 Training, 1.22.1	Within twenty (30) days of contract award and new employee status and every twelve (12) months thereafter	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 13</i> , OPSEC Training, 1.22.2	Within twenty (30) days of contract award and new employee status and every twelve (12) months thereafter	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 14</i> , iWATCH Training, 1.22.3	Within twenty (30) days of contract award and new employee status and every twelve (12) months thereafter	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 15</i> , Proprietary Systems, 4.1.1	As required when the systems require proprietary certification to perform work on those systems	N/A	Memorandum of Agreement, Memorandum of Understanding, and/or Written Statement	COR/KO
<i>Deliverable 16</i> , Equipment Inventory, 5.2.2	Thirty (30) days after requested	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 17</i> , SAM, 5.4.2	Throughout life of contract	N/A	Via website	SAM

<u>Deliverable</u>	<u>Frequency</u>	<u># of Copies</u>	<u>Medium/Format</u>	<u>Submit To</u>
<i>Deliverable 18,</i> Contractor shall document findings during PM that may require further work/DMO to be issued, 5.2.3	Immediate upon findings, at minimum prior to any further work.	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 19,</i> Individual parts costing over € 2500 must be completed by obtaining at least three (3) quotes, 5.3.3	Quotes will be provided to the Contracting Officer via the COR for decision prior to beginning work.	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO
<i>Deliverable 20,</i> The Contractor shall submit all invoices (electronically via WAWF), 5.3.5	Within thirty (30) days from completion of work	One each through WAWF	Electronically using WAWF	COR using WAWF
<i>Deliverable 21,</i> When requested by the COR or KO, the contractor shall prepare and provide reports to include but no limited to: Parts Consumption, Labor Hours, Invoice Costs, Waste Removal, Technical Data, Maintenance Schedules etc. 5.3.6	When requested reports will be provided to the COR/KO within five (5) days.	One each	Electronic Document via email using Microsoft Office or PDF	COR/KO

**TECHNICAL EXHIBIT C (TE-C)
HVAC EQUIPMENT INVENTORY**

UBBS:

AIR HANDLING UNITS (LBK'S)							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M ³ /h	MFR YEAR
90005	DPWNL843.90005 BACK	SYSTEMAIR	AIR HANDLING UNIT (LBK6 STORAGE)	HH COMPACT P 08.08	LBK 6	3996 W (4,0 kw) @ 8 000 m ³ /h	2018
90005	DPWNL843.90005 BACK	SYSTEMAIR	AIR HANDLING UNIT (LBK3 FITNESS)	HH COMPACT P 08.08	LBK 3	3996 W (4,0 kw) @ 8 000 m ³ /h	2018
90005	DPWNL843.90005 BACK	SYSTEMAIR	AIR HANDLING UNIT (LBK4 BOWLING)	HH COMPACT P 09.09	LBK 4	5339 W (5,3 kw) @ 10 125 m ³ /h	2018
90005	DPWNL843.90005 FRONT	SYSTEMAIR	AIR HANDLING UNIT (LBK1 DINING)	HH COMPACT R 08.08	LBK 1	4662 W (4,7 kw) @ 8 000 m ³ /h	2018
90005	DPWNL843.90006 FRONT	SYSTEMAIR	AIR HANDLING UNIT (LBK2 MUP)	HH COMPACT R 08.08	LBK 2	4662 W (4,7 kw) @ 8 000 m ³ /h	2018
90005	DPWNL843.90007 FRONT	SYSTEMAIR	AIR HANDLING UNIT (LBK5 KITCHEN)	HH COMPACT R 06.06	LBK 5	2875 W (2,9 kw) @ 4 500 m ³ /h	2018
90005	DPWNL843.90008 FRONT	SYSTEMAIR	AIR HANDLING UNIT (LBK7 LOUNGE)	HH COMPACT R 07.07	LBK 7	3910 W (3,9 kw) @ 6 125 m ³ /h	2018
90006	DPWNL843.90006.123 2nd FLOOR	SYSTEMAIR	AIR HANDLING UNIT	DV60	1906-0005287159-10	10 080 W (10,1 kw) @ 17 280 m ³ /h	2018
90007	DPWNL84390007	SYSTEMAIR	AIR HANDLING UNIT	DV40	1805-0004794026-10	6300 W (6,3 kw) @ 10 800 m ³ /h	2018
90008	STAIRCASE	SYSTEMAIR	AIR HANDLING UNIT (LBK4)	HH FLEX 14.14	LBK 4	17155m ³	2018
90008	ROOF	SYSTEMAIR	AIR HANDLING UNIT (LBK5)	HH FLEX 13.13	LBK 5		2018
90008	ROOF	SYSTEMAIR	AIR HANDLING UNIT (LBK6)	HH FLEX 10.12	LBK 6		2018
90009	DPWNL843900009	SYSTEMAIR	AIR HANDLING UNIT	TIME 10	0004971386-10	1680 W (1,7 kw) @ 2 880 m ³ /h	2018
90013	DPWNL84390013	SYSTEMAIR	AIR HANDLING UNIT	TIME 10	0004739658-10	1680 W (1,7 kw) @ 2 880 m ³ /h	2018
90014	ROOF	SYSTEMAIR	AIR HANDLING UNIT	DV15	1709-0004483694-10	2520 W (2,5 kw) @ 4 320 m ³ /h	2018

HEAT PUMPS							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M ³ /h	MFR YEAR
90005	TECH ROOM	CARRIER	HEAT PUMP, R-410A, AIR-TO-WATER, 13 TON	30RQSY045B0013-PE-	M2019000076	13.50KgR-410A	2019
90005	TECH ROOM	CARRIER	HEAT PUMP, R-410A, AIR-TO-WATER, 44 TON	30RQSY160B0028-PE-	M2019000075	37kgR-410A	2019
90006	TECH RM 2 ND FLOOR	CARRIER	HEAT PUMP, R-410A, AIR-TO-WATER, 10 TON	30RQSY129B0018-PE-	M2019011518	33kg R-410A	2019
90007	DPWNL84390007	CARRIER	HEAT PUMP, R-410A, AIR-TO-WATER, 17 TON	30RQSY07080014-PE-	M2018010497	18kg R-410A	2018
90008	ROOF	CARRIER	HEAT PUMP, R-410A, AIR-TO-WATER, 64 TON	30RQP-230A0022-PE	M2018015854	R-410A 48Kg	2018
90008	ROOF	CARRIER	HEAT PUMP, R-410A, AIR-TO-WATER, 32 TON	30RQS-120B0084-PE-	M2018016273	R-410A 33Kg	2018

AC UNITS							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M³/h	MFR YEAR
90001	GUARD HOUSE	DAIKIN	AC UNIT	RZAG35A2V1B	J030513	220-240V/50HZ	2024
90001	GUARD HOUSE	MITSUBISHI	AC UNIT	MUZ-AY35VG	3C063584TR	230V/3.5-4.0KW	2023
90001	GUARD HOUSE	MITSUBISHI	AC UNIT	MUZ-AY35VG	3C063569TR	230V/3.5-4.0KW	2023
90002	KIBO UNIT 1	DAIKIN	AC UNIT	RXM50R5V1B	T019447	220-240V/0.6A	2022
90002	KIBO UNIT 2	DAIKIN	AC UNIT	RXM50R5V1B	T027408	220-240V/0.6A	2022
90027	OVERWATCH	DAIKIN	AC UNIT	RXC35BV1B	K026531	220-240V/3.5-4.0KW	2019
T.B.D.	GUARD SHACK 1	MITSUBISHI ELECTRIC	AC UNIT	MUZ-HR35VF	0C085499TR	230V/28W/0.27A	2020
T.B.D.	GUARD SHACK 2	DAIKIN	AC UNIT	RXC35BV1B	K026555	220-240/3.5-4.0KW	2019
90005	OUTSIDE GROUND	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1.4 TON	RXP50M2V1B	J002889	R32/0.9kg	2019
90005	OUTSIDE GROUND	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1.4 TON	RXP50M2V1B	J000916	R32/ 0.9Kg	2019
90005	OUTSIDE GROUND	DAIKIN	AC CASINO	RZASG125M7V1B	1903002	R32/2.6 Kg	2019
90005	OUTSIDE GROUND	PROFROID	CONDENSING UNIT (CO2 REFRIGERATION SYSTEM)	QC MT 50 PK	211745/141/1362	R744	2018
90005	OUTSIDE GROUND	PROFROID	CONDENSING UNIT (CO2 REFRIGERATION SYSTEM)	QQ LT 75 PK	211745/141/1370	R744	2018
90006	OUTSIDE GROUND	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1/2 TON	RXP20MV1B	T006299	R32/0.55kg	2019
90007	ROOF	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1 TON	RXP25L2V1B	J101417	R32/0.65Kg	2018
90008	ROOF SERVER ROOM 248	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1 TON	RXP35L2V1B	J103269	R32/0.7Kg	2018
90008	ROOF SECURITY 118	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1 TON	RXP35L2V1B	J100439	R32/0.7Kg	2018
90008	ROOF SECURITY 118a	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1 TON	RXP35L2V1B	J103268	R32/0.7Kg	2018
90008	ROOF DATA SPACE	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1 TON	RXP35L2V1B	J100437	R32/0.7Kg	2018
90008	ROOF DATA SPACE	TOSHIBA	AC, MULTI SPLIT VRF, 3 TON	RAV-GM1102AT8W-E	7055074501297121	R32/2.40Kg – 11 KW	2025
90009	ROOF	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1 TON	RXC35CV1B	K008496	R32/0.75kg	2020
90013	ROOF	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1.7 TON	RXM60N2V1B	J001471	R32/1.15kg	2018
90013	ROOF	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1.7 TON	RXM60N2V1B	J001498	R32/1.15kg	2018
90013	ROOF	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1.7 TON	RXM60N2V1B	J001496	R32/1.15kg	2018
90013	ROOF	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1.7 TON	RXM60N2V1B	J001499	R32/1.15kg	2018
90013	ROOF	DAIKIN		RZASG100M7Y1B	3214964	R32/2.6KG	2022
90013	ROOF	DAIKIN		RZASG100M7Y1B	3214966	R32/2.6KG	2022
90014	ROOF	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1.4 TON	RXP50K3V1B	J003912	R32/1.4kg	2018
90014	ROOF	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1.4 TON	RXP35K3V1B	J003910	R32/1.4kg	2018
90014	ROOF	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1 TON	RXP50K3V1B	J009809	R32/0.76.4kg	2017

BOILERS (GAS-FIRED)							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M³/h	MFR YEAR
90005	TECH RM 2 ND FLOOR	REMEHA	WATER HEATER, GAS FIRED, 84 KW	QUINTA ACE 90	1830319893520	84.2Kw-89.5Kw	
90005	TECH RM 2 ND FLOOR	REMEHA	WATER HEATER, GAS FIRED, 61.5 KW	QUINTA ACE 65	1831720013730	61.5Kw-65Kw	
90005	TECH RM 2 ND FLOOR	REMEHA	WATER HEATER, GAS FIRED, 61.5 KW	QUINTA ACE 65	1831720013810	61.5Kw-65Kw	
90006	TECH RM 2 ND FLOOR	REMEHA	BOILER, GAS FIRED, 89.5 KW	QUINTA ACE 90	1923921968550	84.2-89.5KW	
90006	TECH RM 2 ND FLOOR	REMEHA	BOILER, GAS FIRED, 61.5 KW	QUINTA ACE 65	1928822395120	61.5-65KW	
90007	TECH RM	REMEHA	BOILER, GAS FIRED, 40.8 KW	QUINTA PRO 45	1817819022760		
90007	TECH RM	REMEHA	BOILER, GAS FIRED, 61.5 KW	QUINTA PRO 65	1817318994870		
90008	BOILER ROOM	REMEHA	CONDENSING GAS BOILER	GAS 610 ECO PRO-570	182 8919777361 L	69-522KW	2018
90008	BOILER ROOM	REMEHA	CONDENSING GAS BOILER	GAS 610 ECO PRO-570	182 8919777362 R	69-522KW	2018
90009	TECH ROOM	REMEHA	BOILER, GAS FIRED, 38.8 KW	CALENTA ACE 40C	1905120793000	34.5 – 38.8 KW	
90013	TECH ROOM	REMEHA	BOILER, GAS FIRED, 38.8 KW	CALENTA ACE 40C	1817719007090	34.5 – 38.8 KW	
90014	TECH ROOM	REMEHA	BOILER, GAS FIRED, 38.8 KW	CALENTA ACE 40C	1724316815970	34.5 – 38.8 KW	

MISCELLANEOUS HVAC EQUIPMENT						
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	QUANTITY	NOTES
90005	TECH ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON 200/1	1	
90005	TECH ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON M 200/3	1	
90005	TECH ROOM	INTERCLIMA	THERMAL BUFFER TANK	PVREH-19/397	1	500L
90005	TECH ROOM	INTERCLIMA	THERMAL BUFFER TANK	INT_VT-V8GEB/1000	1	1000L
90005	TECH ROOM 2 ND FLOOR	GRUNDFOS	CIRCULATION PUMP	MAGNA 3	8	
90005	TECH ROOM LEFT	GRUNDFOS	CIRCULATION PUMP	ALPHA 2	6	LBK'S – BOILER ROOM – STAIRCASE
90005	TECH ROOM 2 ND FLOOR	FLAMCO	EXPANSION VESSEL	FLEXCON 12/1	1	
90005	TECH ROOM RIGHT	GRUNDFOS	CIRCULATION PUMP	ALPHA 2	5	
90005	TECH ROOM RIGHT	GRUNDFOS	CIRCULATION PUMP	MAGNA 3	1	
90006	TECH ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON PREMIUM 80/1	2	6 BAR/1 BAR/ 90°C
90006	TECH ROOM 2 ND FLOOR	FLAMCO	EXPANSION VESSEL	CONTRA-FLEX 18-1	2	3 BAR / 1 BAR / 70°C
90006	TECH ROOM		THERMAL BUFFER TANK		1	INSULATED
90006	TECH ROOM	GRUNDFOS	CIRCULATION PUMP	MAGNA 3	6	
90006	TECH ROOM 2 ND FLOOR	SPIROTECH	DIRT & AIR SEPARATOR	SPIROCOMBI BC065F	1	
90006	TECH ROOM		COMPRESSOR		1	
90007	ROOF	SYSTEM AIR	EXHAUST FAN	MULTIPLE	2	
90007	TECH ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON 50/1	1	
90007	TECH ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON 80/1	1	
90007	TECH ROOM	CORDIVARI	THERMAL BUFFER TANK	500L	1	
90007	TECH ROOM	GRUNDFOS	CIRCULATION PUMP	MAGNA 3	3	
90008	ROOF	SYSTEM AIR	EXHAUST FAN	MULTIPLE	5	

90008	BOILER ROOM	SPIROTECH	VACUUM DEGASSER / MAKE-UP WATER SYS	SPIROVENT SUPERIOR S400	1	100M3/1-4 BAR(G)/ 90°C
90008	BOILER ROOM	SPIROTECH	DIRT & AIR SEPARATOR	SPIROCOMBI BC125F	1	
90008	BOILER ROOM	SPIROTECH	DIRT & AIR SEPARATOR	SPIROCOMBI BC080F	1	
90008	BOILER ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON 18/1	2	
90008	BOILER ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON 300/1	1	
90008	BOILER ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON 200/1	2	
90008	BOILER ROOM	GRUNDFOS	CIRCULATION PUMP	MAGNA 3	11	
90008	BOILER ROOM	GRUNDFOS	CIRCULATION PUMP	TPE SERIES	2	MGE90LC / MGE100LA
90008	BOILER ROOM		THERMAL BUFFER TANK		2	INSULATED
90013	ROOF	SYSTEM AIR	EXHAUST FAN	MULTIPLE	1	
90013	TECH ROOM	REFLEX	EXPANSION VESSEL	35L	1	
90014	TECH ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON 25/05	1	
90014	TECH ROOM		COMPRESSOR	TWINTANK 15 BAR	1	
90014	ROOF	SYSTEM AIR	EXHAUST FAN	MULTIPLE	1	

APS EYG:

AIR HANDLING UNITS (LBK'S)							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M³/h	MFR YEAR
60009	TECH ROOM LEFT	VERHULST	AIR HANDLING UNIT, LBK1 LEFT	VK-7.5D	G92886	8.33m³/s	1985
60009	TECH ROOM RIGHT	VERHULST	AIR HANDLING UNIT, LBK2 RIGHT	VK-7.5D	G92886	8.33m³/s	1985
60011	COMMS ROOM	DAIKIN	ENERGY RECOVERY UNIT, HRU1	VAM500J7VEB	2106740	500 M3/H	2021
60011	COMMS ROOM	DAIKIN	ENERGY RECOVERY UNIT, HRU2	VAM800J7VEB	2104785	500 M3/H	2021
60011	ARMS ROOM	DAIKIN	ENERGY RECOVERY UNIT, HRU3	VAM650J7VEB	2104159	500 M3/H	2021
60011	ARMS ROOM	DAIKIN	ENERGY RECOVERY UNIT, HRU4	VAM650J7VEB		500 M3/H	2021
60012	MR 2nd FLOOR OFFICE	VERHULST	AIR HANDLING UNIT, LBK3	VK-2.1-B	order nr 397810 (LBK 3)	2.08 m³/s	1985
60012	MR 2nd FLOOR OFFICE	VERHULST	AIR HANDLING UNIT, LBK2	VK-4-B	order nr K92810 (LBK 2)	3.89 m³/s	1985
60012	MR 2nd FLOOR OFFICE	VERHULST	AIR HANDLING UNIT, LBK1	VK-1.5B	order K92810 (LBK 1)	1.25 m³/s	1985
60012	MR 2nd FLOOR MAINT.	VERHULST	AIR HANDLING UNIT, LBK6	VK 4B	order K92811 (LBK 6)	3.61 m³/s	1985
60012	MR 2nd FLOOR MAINT.	VERHULST	AIR HANDLING UNIT, LBK5	VK 2-1-B	order K92812 (LBK 5)	2.22m³/s	1985
60012	MR 2nd FLOOR MAINT.	VERHULST	AIR HANDLING UNIT, LBK7	VK3B	order K92813 (LBK 7)	2.03 m³/s	1985
60012	ROOF CRSP	VERHULST	AIR HANDLING UNIT, LOKAAL 34	VK-5-D-B-U	order nr K92810 (CRSP)	5.56 m³/s	1985
60012	ROOF RIGHT	DUTCH BLOWER	VENT FAN, CENTRIFUGAL, EF24	BDA76	C1482-1	1000m³/h	2001
60028	BACKSIDE BLDG	WOLF	AIR HANDLING UNIT, BLDG 60028	KG Top 2510	2205	6500M3/H	2025

HEAT PUMPS							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M³/h	MFR YEAR
60028	TECH ROOM MAIN ENTRANCE	GALLETTI	HEAT PUMP, WATER-TO-WATER, 4 PIPE SYSTEM	30LEP042PL_F1	HF2502132020	R410A/470KW	2025

AC UNITS							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M³/h	MFR YEAR
60009	ROOF	LG	MINI SPLIT, DUCTLESS, AC1	-	-	-	-
60009	ROOF	LG	MINI SPLIT, DUCTLESS, AC2	-	-	-	-
60011	OUTSIDE GROUND	DAIKIN	CONDENSING UNIT, SPLIT SYSTEM	RXYQ10U7Y1B	3106734		2021
60011	WALL MOUNTED	DAIKIN	MINI SPLIT, DUCTLESS, AC1	RXM50R2V1B	J039720	R32/ 1.15KG	2021
60011	WALL MOUNTED	DAIKIN	MINI SPLIT, DUCTLESS, AC2	RXM50R2V1B	J039718	R32/ 1.15KG	2021
60012	ROOF	LG	MINI SPLIT, DUCTLESS, AC1	LS-K1862RL	TAG: 900095946	R410A 1.30Kg	-
60012	ROOF	FUJI	MINI SPLIT, DUCTLESS, AC2	ROR12LGC	E006183	R410A0.08Kg	-
60012	ROOF	mitsubishi	MINI SPLIT, DUCTLESS, AC3	T10410108-23RD2	TAG: 900095942	R407/ 1.07 kg	-

60012	ROOF	DAIKIN	MINI SPLIT, DUCTLESS, AC4	RXM35N2V1B9	J043586	R32/0.76 kg	2019
60012	ROOF	DAIKIN	MINI SPLIT, DUCTLESS, AC5	T10410108-23RD1	TAG: 900095943	R407/ 1.07 kg	-
60012	ROOF	DAIKIN	MINI SPLIT, DUCTLESS, AC6	RXM35N2V1B9	J043624	R32 / 0.76kg	2019
60012	ROOF	DAIKIN	MINI SPLIT, DUCTLESS, AC7	RXM50N2V1B	J000912	R321.15Kg	2018
60012	ROOF	DAIKIN	MINI SPLIT, DUCTLESS, AC8	RXM50N2V1B	J002106	R32 / 1.15Kg	2018
60012	ROOF	LG	MINI SPLIT, DUCTLESS, AC9	LS-L1266ML	TAG: 900095944	R410A1.05Kg	-
60012	ROOF	LG	MINI SPLIT, DUCTLESS, AC10	LS-L1266ML	TAG: 900095945	R410A 0.8Kg	-
60012	ROOF	FUJITSU	MINI SPLIT, DUCTLESS, AC11	AOYG18KLTA	T010878	R32 / 0.85Kg	2019
60012	ROOF	DAIKIN	AC UNIT, SPLIT SYSTEM, AC12	RXYSQ6T8YB	1700414	R410A 3.6Kg	2017
60012	ROOF	DAIKIN	MINI SPLIT, DUCTLESS, AC13	SRC50ZS-W	368701089CE	R32 / 1.05Kg	-
60012	ROOF	DAIKIN	AC UNIT	INVERTER R32	308993.036	R32/1.15kg	2018
60012	ROOF	DAIKIN	AC UNIT	INVERTER R33	303993.037	R32/1.15kg	2018
60023	OUTSIDE SERVER ROOM	DAIKIN	MINI SPLIT, DUCTLESS, AC1	RXM35M3V1B9	J055778	R32 / 0.76Kg	2018
60023	BUILDING ENTRANCE	DAIKIN	AC UNIT	RXYSQ8TMY1B	F030409	R410A/ 5.5KG	2021
60025	ROOF	MITSUBISHI ELECTRIC	AC UNIT	MUZ-AY25VG	4C029453TR	R32/0.55KG	2024
60025	ROOF	MITSUBISHI ELECTRIC	AC UNIT	MUZ-AY25VG	4C029469TR	R32/0.55KG	2024
60025	WALL MOUNTED BACK	DAIKIN	AC UNIT	RXM42M3V1B9	J008932	R32/1.30KG	2018
60025	WALL MOUNTED	LG	AC UNIT	S09EQUA3	006TKXL29746	R32 0.7Kg	-

BOILERS (GAS-FIRED)

BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M³/h	MFR YEAR
60012	BOILER ROOM	REMEHA	BOILER, GAS-FIRED, B1, 5000 MBH	B23	2003Z31512775	1630 kW	2003
60012	BOILER ROOM	REMEHA	BOILER, GAS-FIRED, B2, 5000 MBH	B23	2004Z41512787	1630 kW	2004
60012	BOILER ROOM	MONARCH	BURNER 1	WM-L20/3-A	40398760	2000KW	2017
60012	BOILER ROOM	MONARCH	BURNER 2	WM-L20/3-A	40398759	2000KW	2017
60019	HVAC CONTAINER	VISSMANN	BOILER, GAS FIRED, 88KW	VITOPLEX 100 / SX1	7324727300590104	88KW	2003
60023	TECH ROOM	REMEHA	BOILER, GAS FIRED, 25KW	CALENTA 28C	1623514291120	25KW	

DEHUMIDIFIERS

BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M³/h	MFR YEAR
60001	TECHNICAL ROOM	INDUSTRIAL AIR	IAC ADSORPTION DRYER	ST-5500 MASTER	20121840	54KW/1270M3/H	2020
60001	TECHNICAL ROOM	INDUSTRIAL AIR	IAC ADSORPTION DRYER	ST-5500 SLAVE	20121839	54KW/1270M3/H	2020
60002	TECHNICAL ROOM	INDUSTRIAL AIR	IAC ADSORPTION DRYER	ST-5500 MASTER	20121842	54KW/1270M3/H	2020
60002	TECHNICAL ROOM	INDUSTRIAL AIR	IAC ADSORPTION DRYER	ST-5500 SLAVE	20121843	54KW/1270M3/H	2020
60003	TECHNICAL ROOM	INDUSTRIAL AIR	IAC ADSORPTION DRYER	ST-5500 MASTER	20121838	54KW/1270M3/H	2020

60003	TECHNICAL ROOM	INDUSTRIAL AIR	IAC ADSORPTION DRYER	ST-5500 SLAVE	20121841	54KW/1270M3/H	2020
60004	TECHNICAL ROOM	MUNTERS	DEHUMIDIFIER, 112 LB/HR	MX ² 55E	1847013056741964	40.1KW	2018
60004	TECHNICAL ROOM	MUNTERS	DEHUMIDIFIER, 112 LB/HR	MX ² 55E	1846013056741960	40.1KW	2018
60005	TECHNICAL ROOM	MUNTERS	DEHUMIDIFIER, 112 LB/HR	MX ² 55E	1846013056741962	40.1KW	2018
60005	TECHNICAL ROOM	MUNTERS	DEHUMIDIFIER, 112 LB/HR	MX ² 55E	1846013056741958	40.1KW	2018
60006	TECHNICAL ROOM	MUNTERS	DEHUMIDIFIER, 112 LB/HR	MX ² 55E	1846013056741961	40.1KW	2018
60006	TECHNICAL ROOM	MUNTERS	DEHUMIDIFIER, 112 LB/HR	MX ² 55E	1846013056741959	40.1KW	2018
60007	TECHNICAL ROOM	MUNTERS	DEHUMIDIFIER, 112 LB/HR	MX ² 55E	1846013056741957	40.1KW	2018
60007	TECHNICAL ROOM	MUNTERS	DEHUMIDIFIER, 112 LB/HR	MX ² 55E	1847013056741963	40.1KW	2018
60008	TECHNICAL ROOM	INDUSTRIAL AIR	IAC ADSORPTION DRYER	ST-5500 MASTER	230706	54KW/1270M3/H	2023
60008	TECHNICAL ROOM	INDUSTRIAL AIR	IAC ADSORPTION DRYER	ST-5500 SLAVE	230707	54KW/1270M3/H	2023

MISCELLANEOUS HVAC EQUIPMENT						
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	QUANTITY	NOTES
60009	TECH ROOM LEFT	GRUNDFOS	CIRCULATION PUMP	UMS 40-30 F06	1	
60009	TECH ROOM RIGHT	GRUNDFOS	CIRCULATION PUMP	UPS 32-60F	1	
60009	TECH ROOM RIGHT	GRUNDFOS	CIRCULATION PUMP	MAGNA 1	1	
60012	MR 2ND FLOOR OFFICE	GRUNDFOS	CIRCULATION PUMP	UPS 25-80 180	1	230V/50HZ
60012	MR 2ND FLOOR OFFICE	GRUNDFOS	CIRCULATION PUMP	UPS 25-50 180	1	230V/50HZ
60012	MR 2ND FLOOR OFFICE	GRUNDFOS	CIRCULATION PUMP	ALPHA 2	1	
60012	MR 2ND FLOOR MAINT	GRUNDFOS	CIRCULATION PUMP	UMS 40-30 F06	1	220V/50HZ
60012	MR 2ND FLOOR MAINT	GRUNDFOS	CIRCULATION PUMP	UPS 40-30 F06	1	220V/50HZ
60012	MR 2ND FLOOR MAINT	GRUNDFOS	CIRCULATION PUMP	UP 40-50 F250	2	400V/50HZ
60012	MR 2ND FLOOR MAINT	WILO	CIRCULATION PUMP	TOP SERIES	1	
60012	BOILER ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON M-K	2	1600L/ 6 bar
60012	BOILER ROOM	GRUNDFOS	CIRCULATION PUMP	UPS 40-120 F06	1	380V/50HZ
60012	BOILER ROOM	GRUNDFOS	CIRCULATION PUMP	UMS 65-60 F06	1	380V/50HZ
60012	BOILER ROOM	GRUNDFOS	CIRCULATION PUMP	UP 32-80 180	1	400V/50HZ
60012	BOILER ROOM	GRUNDFOS	CIRCULATION PUMP	UPS 50-120/F	1	400V/50HZ
60012	BOILER ROOM	GRUNDFOS	CIRCULATION PUMP	UP 40-75R	2	380V/50HZ
60012	BOILER ROOM	GRUNDFOS	CIRCULATION PUMP	MAGNA 1	1	
60012	BOILER ROOM	GRUNDFOS	CIRCULATION PUMP	MAGNA 3	1	
60012	ROOF	SYSTEM AIR	EXHAUST FAN	MULTIPLE	26	
60019	HVAC CONTAINER	A.J. DIJKMAN	FUEL TANK		1	2M3

JFC (H-310 & CYSS)

AIR HANDLING UNITS (LBK'S)							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M ³ /h	MFR YEAR
H-310	ROOF	LIKO					
H-602	SAC TECH ROOM	TROGES AIRSET	AIR HANDLING UNIT (LBK)	Type 18.09XL	12789-0001/2012	18800m ³ /h	2012

HEATPUMPS							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M ³ /h	MFR YEAR
H-602	TECH ROOM	TECHNECO ENERGIESYSTEMEN	HEAT PUMP, 55 GAL (GROUND SOURCE/BRINE TO WATER)	VULCAN 43AHR	PSO43B112158001	R407C/44.4KW	2015

AC & REFRIGERATION UNITS							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M ³ /h	MFR YEAR
H-310	ROOF	UNIFLAIR	REFRIGERATION CONDENSER (AIR COOLED)	30CA0P38P4	066441	R410A / 45 BAR	2021
H-310	ROOF	UNIFLAIR	REFRIGERATION CONDENSER (AIR COOLED)	30CA0P38P4	066442	R410A / 45 BAR	2021
H-310	ROOF	MODINE CIS ITALY	REFRIGERATION EVAPORATOR (AIR COOLED)	CAP036IP	CA202100481	R410A	2021
H-310	ROOF	MODINE CIS ITALY	REFRIGERATION EVAPORATOR (AIR COOLED)	CAP036IP	CA202100482	R410A	2021
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 6.4 TON	RXYQ8U7Y1B	3205490	R410A, 5.9kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 6.4 TON	RXYQ8U7Y1B	3209118	R410A, 6.3kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 6.4 TON	RXYQ8U7Y1B	3205493	R410A, 5.9kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 6.4 TON	RXYQ8U7Y1B	3209097	R410A, 6.3kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 6.4 TON	RXYQ8U7Y1B	3205489	R410A, 6.3kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A	RXYQ12U7Y1B	3209096	R410A, 6.3kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 11.3 TON	REYQ14U7Y1B	5203245	R410A, 11.8kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 12.79 TON	REYQ16U7Y1B	5205125	R410A, 11.8kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 12.79 TON	REYQ16U7Y1B	5205130	R410A, 11.8kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 6.4 TON	ERQ250A7W1B	8213685	R410A, 8.4kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 7.96 TON	ERQ250A7W1B	8213687	R410A, 8.4kg	2022
H-602	AC YARD	DAIKIN	CONDENSER, DX, R-410A, 6.4 TON	RXYQ8U7Y1B	3205494	R410A, 5.9kg	2022
H-602	WALL MOUNTED FRONT BLDG	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1/2 TON	RXM20M3V3B9	J001899	R32 0.76 Kg	2017
H-602	WALL MOUNTED FRONT BLDG	HAIER	AC, MINI SPLIT SYSTEM, R-32, 1.4 TON	1U50MEEFRA	0011501458	R32 0.90Kg	
H-602	WALL MOUNTED	DAIKIN	AC, MINI SPLIT SYSTEM, R-32, 1/2 TON	RXM35N2V1B9	J001631	R32 0.76 Kg	

BOILERS (GAS-FIRED)							
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	SERIAL #	Watts/KW/M ² /h	MFR YEAR
H-602	TECH ROOM	VISSMANN	CONDENSING GAS-FIRED BOILER (CLOSED PRESSURIZED SYSTEM)	VITOCROSSAL 200 CM2	7436867202641100	105 – 115 KW / 4 BAR @ 110 °C	2012

MISCELLANEOUS HVAC EQUIPMENT						
BLDG	LOCATION	MFR	SYSTEM TYPE	MODEL #	QUANTITY	NOTES
H-602	TECH ROOM	REFLEX	EXPANSION VESSEL	REFLEX N 200	1	200L
H-602	TECH ROOM	UNKOWN	THERMAL BUFFER TANK	UNKOWN	1	1000L INSULATED
H-602	TECH ROOM	UNKOWN	AIR SEPARATOR, 3 INCH	DN80	1	INSULATED
H-602	TECH ROOM	REFLEX	EXPANSION VESSEL	REFLEX N	1	18L
H-602	TECH ROOM	FLAMCO	EXPANSION VESSEL	FLEXCON 18/05	1	18L
H-602	TECH ROOM	WILO	CIRCULATION PUMP	TOP-S 50/4	1	230V/50HZ
H-602	TECH ROOM	WILO	CIRCULATION PUMP	TOP-S 25/2	2	230V/50HZ
H-602	TECH ROOM	WILO	CIRCULATION PUMP	TOP-S 40/4	1	230V/50HZ
H-602	TECH ROOM	WILO	CIRCULATION PUMP	IL SERIES	1	(LIKELY IL 65/150-11)
H-602	TECH ROOM	REFLEX	AUTOMATIC FILLING UNIT	SERVITEC 35 MHR	1	3.5M3/H / 3 BAR
H-602	ROOF	UNKOWN	EXHAUST FAN	MULTIPLE	2	1500 CFM

**ATTACHMENT 1
ESTIMATED WORKLOAD DATA**

ITEM	NAME	ESTIMATED QUANTITY	
		Hours	
1.	Project Management/Admin	Hours	96
2.	Boiler and Combustion Systems	Hours	114
3.	Hydronic Miscellaneous Systems	Hours	28
4.	Refrigeration and Air-Conditioning Systems	Hours	257
5.	Air Side Systems	Hours	202
6.	Dehumidifiers	Hours	96
7.	Inventory and labeling	Hours	60
8.	Site Visits	Hours	200

ATTACHMENT 2
HISTORICAL WORKLOAD DATA

ITEM	NAME	ESTIMATED QUANTITY	
N/A	N/A	N/A	N/A

DRAFT

ATTACHMENT 3
GOVERNMENT FURNISHED PROPERTY (GFP)

RESERVED

DRAFT